

The Machine in the Market: Algorithmic Intermediation and Price Discovery in the Housing Market*

Saani Rawat
Marquette University

June 25, 2026

Preliminary and incomplete — please do not cite without permission

Abstract

Between 2014 and 2021, iBuyers became the first intermediaries to quote firm, capital-backed prices generated by statistical models at scale in the largest decentralized asset market. I ask whether the machine’s quotes sharpened price discovery for households who never traded with it. A model of algorithmic intermediation with adverse selection predicts that the machine sorts into homogeneous segments, that forecast-error variance taxes its spread (so the highest-error algorithm exits first when home prices become unpredictable), and that its public purchase prices act as signals that compress price dispersion among third-party trades—by an amount that scales with the machine’s tiny share of comparable sales. I take the model to 42 million deed records across 22 states, using the abrupt November 2021 shutdown of Zillow Offers, with ZIPs exposed to the surviving machines as controls. The cleanly identified result is a precise null: when the machine’s quotes vanished, within-ZIP price dispersion did not measurably rise. The effect is statistically zero with a clean pre-period placebo, and the bound is economically tight—at the machine’s reach

*Saani Rawat, Marquette University. Email: saani.rawat@marquette.edu. I thank ... for helpful comments. All errors are my own.

of under two percent of comparable-sale windows, the data cap the externality below one percent of baseline dispersion, sharp enough to reject the model’s most aggressive calibrations and to confirm its central message: an algorithmic dealer at this scale is too small a part of the comparable-sale record to move price discovery. Household transaction volume also fell sharply in exposed ZIPs, but exposed markets were on a differential downward path before the shutdown, so I bound rather than assert a causal liquidity effect. The lesson is not that the machine’s information was worthless but that, at the scale the industry reached, what the market measurably lost when the algorithm died was a balance sheet, not a signal—and the advisory Zestimate, the signal, never went offline.

JEL classification: D47, D82, D83, G14, L85, R31

Keywords: algorithmic intermediation, iBuyers, price discovery, housing search, adverse selection, AVM

1 Introduction

The United States housing stock is worth roughly \$50 trillion, and the market that prices it has no market maker. Every house is different. The typical owner trades once a decade. The information that determines value—the slope of the lot, the smell in the basement, the neighbor’s plans—is dispersed across millions of households and revealed only through bilateral search and bargaining. Prices in this market aggregate information slowly and badly: house price changes are forecastable (Case and Shiller, 1989), momentum persists for years (Glaeser and Nathanson, 2017; Guren, 2018), sellers learn about demand only by listing and waiting (Anenberg, 2016), and the idiosyncratic component of individual house returns is enormous (Giacoletti, 2021). The microstructure of this market—search, bargaining, brokerage—has been studied for decades (Genesove and Han, 2012; Han and Strange, 2015; Krainer, 2001; Wheaton, 1990). What it never had was an agent willing to post a firm price.

Between 2014 and 2021 it briefly got one. iBuyers—Opendoor, Offerpad, Zillow Offers, RedfinNow—deployed automated valuation models (AVMs) that generated instant, binding cash offers on owner-occupied homes, backed by the firms’ own balance sheets. A homeowner entered an address, received a price within days, and could close in a week. This is not a Zestimate. An advisory estimate is a newsletter; an iBuyer quote is a dealer’s bid. The

firm commits capital against its forecast, takes the house into inventory, and resells it. At peak, the machines were large: iBuyers bought roughly 70,000 homes in 2021, about 1.3 percent of all U.S. home purchases, and in Phoenix—the industry’s birthplace and deepest market—12.1 percent of homes sold in the third quarter of 2021 moved through an iBuyer. For the first time in the history of the largest consumer asset market, a statistical model quoted executable prices at scale.

This paper asks what the machine’s presence did to the market around it. The question is the one [Hendershott et al. \(2011\)](#) asked of algorithmic trading in equities—does the algorithm improve liquidity and make quotes more informative?—transplanted to a decentralized search market where the answer need not be the same. In a centralized limit-order book, algorithmic liquidity provision narrows spreads and speeds price discovery ([Brogaard et al., 2014](#); [Hendershott et al., 2011](#)), though fast trading can also impose adverse-selection externalities ([Biais et al., 2015](#)). Housing is the opposite environment: no consolidated tape, no order book, heterogeneous assets, and counterparties who know things about their own house that no model can see ([Kurlat and Stroebel, 2015](#); [Stroebel, 2016](#)). Whether a machine that quotes firm prices against better-informed homeowners sharpens or distorts price discovery for third parties is a genuinely open question. Theory cuts both ways: a dealer’s quote impounds information into public view ([Glosten and Milgrom, 1985](#); [Kyle, 1985](#)), but an algorithm that free-rides on past transaction prices inherits the [Grossman and Stiglitz \(1980\)](#) limit, and [Akerlof \(1970\)](#) selection bounds what it can touch.

The question has a second half that makes it answerable. On November 2, 2021, Zillow announced that it was shutting down Zillow Offers—then the second-largest iBuyer, operating in 25 metropolitan areas—after a quarter in which it bought 9,680 homes, sold 3,032, and wrote down its inventory by \$304 million. CEO Rich Barton’s explanation was a statement about forecasting, not about housing demand: “We’ve determined the unpredictability in forecasting home prices far exceeds what we anticipated and continuing to scale Zillow Offers would result in too much earnings and balance-sheet volatility.” The shutdown was national, abrupt, and firm-specific. Opendoor and Offerpad kept buying in the same ZIP codes. And critically, Zillow’s advisory signal—the Zestimate, displayed on every listing in America—stayed online. The exit removed the firm, capital-backed quote while leaving the free advisory estimate in place. It is the cleanest separation we are likely to get between the two channels through which an algorithm can inform a

market, and to my knowledge no paper has used it.

I make three contributions. The first is a theory of algorithmic intermediation in a decentralized market. A machine runs an AVM with bounded forecast error and posts a firm bid, net of a spread, to sellers who privately observe their house’s idiosyncratic value. Sellers differ in urgency; the machine resells at fundamental value. The model delivers four results. First, the machine optimally prices so that only urgent sellers—movers—accept, and its profit falls with segment heterogeneity: with any fixed cost of operating a market segment, the machine sorts into homogeneous, machine-readable segments. Second, forecast-error variance is a pure tax on the spread, and the machine faces an operational winner’s curse—it buys the most precisely when it overprices—so a regime shift that makes prices harder to forecast removes the highest-error algorithm first while precise competitors stay. That is the 2021 collapse in one inequality. Third, because the machine’s purchase prices are public deed records, de-spreading them yields a common-value signal for everyone else: machine presence compresses the dispersion of household-to-household prices around fundamentals, and exit raises dispersion in proportion to the machine’s pre-exit share. Fourth, intermediation mechanically adds recorded volume (each intermediated house generates two deeds) and absorbs urgent sellers who would otherwise transact at fire-sale discounts, thinning the left tail of the price distribution; exit reverses both.

The second contribution is the deed evidence on entry. I build a ZIP-by-quarter panel from CoreLogic deed records covering 22 iBuyer-active states from 2010 through mid-2024, identify 376,256 iBuyer-linked deeds by matching buyer and seller names against entity aliases verified in SEC subsidiary filings, and trace the staggered entry of iBuyers across metropolitan areas with interaction-weighted event studies (Sun and Abraham, 2021). Entry is not random—iBuyers chose flat, homogeneous, liquid Sunbelt markets, exactly as the model’s sorting result predicts and as prior work documents (Harrison et al., 2024; Seiler and Yang, 2023)—and the deed panel says so bluntly: joint pre-trend tests reject for most outcomes, so I read the entry event studies as descriptive and place no causal weight on them. The one outcome whose pre-trends are clean (the robust, IQR-based dispersion measure, joint $p = 0.77$) falls by 0.4 log points after entry ($p = 0.08$), the model-consistent sign, with the same sign in all five robustness variants—though as one outcome surviving of five tested, I treat even this as suggestive rather than confirmatory. The entry design’s visible selection is itself a finding—it is Proposition 1 in the raw data—and it is why the paper’s identification

leans entirely on the exit.

The third contribution is the exit experiment. I measure each ZIP code’s exposure to the shutdown as Zillow’s share of transactions over 2018–2021 and compare high-exposure ZIPs to ZIPs with comparable Opendoor and Offerpad exposure, within metro-by-quarter cells. The Opendoor control absorbs whatever made a ZIP attractive to algorithmic buyers in the first place and the 2022 rate shock that hit iBuyer-heavy Sunbelt markets, because Opendoor-exposed ZIPs took that shock too—their machine just kept operating. The identifying assumption is that, conditional on metro-quarter fixed effects and overall iBuyer attractiveness, the incidence of Zillow’s forecasting-driven collapse was uncorrelated with ZIP-level counterfactual trends. The price-discovery outcomes meet that bar: within-ZIP dispersion of household prices rose by a statistically indistinguishable 0.84 log points per percentage point of Zillow exposure (s.e. 1.06), the composition-free IQR measure moved -0.20 (s.e. 0.63), and—crucially—neither shows a differential pre-trend, so the null is identified, not an artifact. The bound is economically tight: the data cap the information externality below about 0.6 percent of baseline dispersion, sharp enough to reject the model’s most aggressive calibrations, while the model fed Zillow’s own forecast-error dispersion predicts an effect comfortably inside it. The reading is not that the machine’s quotes were uninformative in principle but that, reaching under two percent of comparable-sale windows, they were too small a part of the record to move segment-level price discovery—and their removal did not. Household transaction volume *also* fell sharply in exposed ZIPs (a point estimate of -10 log points per exposure point, $p < 0.01$ by permutation and wild bootstrap), but here identification is weaker: exposed ZIPs were on a differential downward volume path before the shutdown, the largest declines arrive in 2023, and a minor second exit (RedfinNow, 2022) does not replicate the pattern. I therefore present the volume decline as consistent with a large liquidity withdrawal but confounded by Zillow’s selection into cooling markets, and bound rather than assert its causal magnitude.

These results speak to three literatures. The closest paper is [Buchak et al. \(2025\)](#), who ask why intermediating houses is so difficult: iBuyers buy at a 3.1 percentage-point discount, earn roughly a 5 percent gross spread, transact on algorithmizable information, and retreat from segments where pricing errors invite adverse selection. Their question is the dealer’s—is machine intermediation profitable, and what caps its scale? Mine is the market’s: does the machine change the quality of price discovery for the people who

never trade with it? The distinction matters because the externality, not the P&L, is what determines whether algorithmic intermediation is worth having. Reduced-form work on iBuyer entry finds price-level and volume effects (Harrison et al., 2024; Seiler and Yang, 2023); I focus instead on second moments—dispersion, tails, informativeness—which is where information lives. Relative to the literature on AVMs as advisory signals—the Zestimate reduces buyer-seller uncertainty (Fu et al., 2025), moves transaction prices (Yu, 2020), and can propagate its own errors through feedback loops (Fu et al., 2022)—the iBuyer quote is executable and capital-backed, and the exit experiment isolates precisely that difference, because the Zestimate never went away. And relative to the industrial-organization literature on algorithmic pricing, where sellers’ algorithms soften competition in gasoline and rental markets (Assad et al., 2024; Calder-Wang and Kim, 2026; Calvano et al., 2020) and machine-learning prediction redistributes surplus in credit (Fuster et al., 2022), the algorithm here is not pricing its owner’s inventory—it is making a market, with its own capital at risk, in everyone else’s.

The housing-search literature supplies the reason to expect any of this to matter. Prices in decentralized housing markets are anchors built from comparables (Guren, 2018), search is segmented (Piazzesi et al., 2020), liquidity is priced in both time-on-market and dispersion (Jiang et al., 2024), and thick-market externalities amplify shocks (Ngai and Tenreyro, 2014). A public, firm, algorithmic quote is a new kind of signal that this literature has never had quasi-experimental variation in. This paper supplies that variation in both directions—arrival and abrupt removal—and finds that the machine’s liquidity died loudly while its information died quietly: the first is visible in deed volume the quarter the algorithm stopped, the second is too small at a one-to-two-percent transaction share to rise above the noise, exactly as the model’s comp-window arithmetic says it must be.

It is worth stating plainly what this paper does and does not show. It *does* show, with a clean design and a tight bound, that the machine’s exit did not measurably degrade price discovery for the households around it—a null the model rationalizes at the machine’s one-to-two-percent reach and the data are sharp enough to make economically informative. It *does not* show that algorithmic intermediation withdrew first-order liquidity: household volume fell sharply in exposed markets, but those markets were already cooling relative to controls before the shutdown, so I document the decline without certifying its cause, and I am explicit that the clean result is the in-

formation null, not the volume number. It does not identify the entry effects causally—iBuyers chose their markets, the pre-trends say so, and I treat the entry evidence as descriptive. And it speaks to the externalities of algorithmic market-making at the modest scale the industry actually reached, not at the scale a future deployment might. What survives all the caveats is a model of an algorithmic dealer with adverse selection that predicts where the machine operated, why the high-forecast-error firm was the one that failed, and how small its price-discovery footprint should be—together with the cleanest evidence to date that, at this scale, the footprint was indeed small.

The rest of the paper proceeds as follows. Section 2 describes the institutions: how an instant offer works, the entry timeline, and the anatomy of the November 2021 collapse. Section 3 presents the model and derives the four testable predictions. Section 4 describes the deed data and the construction of outcomes. Section 5 lays out the two research designs. Section 6 reports results, Section 7 interprets magnitudes and welfare, and Section 8 concludes.

2 Institutional Background

2.1 How an instant offer works

An iBuyer transaction begins with an address. The homeowner submits it through the firm’s website; the firm’s automated valuation model produces an estimate of the home’s market value from recorded characteristics, recent comparable sales, and local price indices; and the firm returns a firm cash offer, typically within 24 to 48 hours, valid for roughly a week. The offer equals the model’s estimate minus a service charge—a posted fee plus, in practice, a purchase discount that compensates the firm for holding risk, renovation, and the lemons it cannot see. Buchak et al. (2025) estimate that iBuyers purchase at a 3.1 percentage-point discount relative to comparable listed homes and earn an average gross spread of about 5 percent. If the seller accepts, an inspection adjusts the price for verifiable defects, the firm closes in days on the seller’s calendar, takes title through a dedicated subsidiary, performs light renovation, and relists. Roughly half of inventory resells within three months (Buchak et al., 2025). Two features of this institution drive everything in this paper. First, the quote is *firm*: the algorithm’s owner commits capital against its own forecast, which an advisory estimate never does. Second, both legs of the trade—the machine’s purchase and its resale—

are recorded deeds, public information available to every appraiser, agent, and AVM in the county.

The economics of the business are a race between two numbers: the liquidity and convenience value the firm delivers to urgent sellers, which it harvests through the spread, and the valuation error of its model, which better-informed homeowners harvest through selective acceptance. A homeowner who knows the house is worth less than the quote accepts; one who knows it is worth more lists on the open market. [Buchak et al. \(2025\)](#) show that iBuyers transact almost exclusively on algorithmizable information—observables plus ZIP-quarter effects explain over 80 percent of price variation in their purchases, against 68 percent in the broader market—and that they avoid segments where algorithmic pricing errors are large. Adverse selection is not an afterthought in this industry; it is the binding constraint, and the model in Section 3 puts it at the center.

2.2 Entry, expansion, and scale

Opendoor invented the category. Founded in March 2014, it bought its first home in Phoenix in December 2014—the flat, tract-built, stucco-homogeneous market that would become the industry’s laboratory. Offerpad followed in 2015, also headquartered in the Phoenix metro. Redfin began experimenting with RedfinNow in Southern California in 2017. Zillow, the dominant consumer real-estate platform, entered last and loudest: Zillow Offers launched in Phoenix in April 2018 and expanded to 25 metropolitan areas over the following 29 months, reaching its 25th and final market, Jacksonville, on September 8, 2020.¹ All four firms paused purchases in March 2020 as COVID-19 froze the housing market—Zillow suspended buying in all markets on March 23, 2020—and all four resumed by August 2020 into the strongest sellers’ market in modern history.

Scale arrived in 2021. iBuyers collectively purchased about 70,000 homes that year, roughly 1.3 percent of all U.S. home purchases and double the previous record. The aggregate number understates local depth, because the machines concentrated geographically exactly as adverse-selection logic predicts: in the third quarter of 2021, 12.1 percent of homes sold in Phoenix

¹Zillow announced the move into direct buying on April 12, 2018 (SEC Form 8-K, Ex-99.1). Expansion dates per company press releases: Las Vegas (June 2018), Atlanta (September 2018), Denver (October 2018), Portland (July 2019), Austin (September 2019), Cincinnati (March 2, 2020, the 24th market), Jacksonville (September 8, 2020, the 25th).

transacted through an iBuyer. Figure 1 plots the time series of iBuyer purchases in my deed data; Figure 2 shows the cross-sectional footprint. The machines bought where houses are young, similar, and easy to price—and within those metros, in the ZIP codes where the housing stock is most homogeneous.

2.3 The collapse of Zillow Offers

Zillow’s 2021 is the natural experiment, so its anatomy matters. Two institutional details set it up. First, in February 2021 Zillow began presenting the Zestimate itself as an initial cash offer in eligible markets—fusing its advisory signal to its dealer quote. Second, on June 15, 2021, Zillow launched the “neural Zestimate,” reporting a national median error rate of 1.9 percent for *on-market* homes but 6.9 percent for *off-market* homes. That asymmetry is the heart of the matter: an instant offer prices an off-market home, sight unseen, precisely where the algorithm’s error is 3.6 times larger. The machine’s public accuracy statistics advertised its strength exactly where its dealer business did not operate.

Through 2021 Zillow scaled aggressively into a decelerating market. Reporting by Insider in November 2021 described an internal initiative, “Project Ketchup,” launched in early 2021 to catch up with Opendoor: employees said it led Zillow to offer prices well above what its algorithm and analysts picked as market value and to buy too many homes in the summer of 2021. In the third quarter of 2021, Zillow Offers purchased 9,680 homes and sold 3,032. On October 17–18, Bloomberg reported that Zillow had stopped signing new purchase contracts for the remainder of 2021, which the company attributed to renovation and closing backlogs. On November 1, Bloomberg reported Zillow was shopping roughly 7,000 homes to institutional investors for about \$2.8 billion. On November 2, 2021, alongside third-quarter results, Zillow announced it would wind down Zillow Offers entirely, taking a \$304 million inventory write-down—the result, the company said, of “purchasing homes in Q3 at higher prices than the company’s current estimates of future selling prices”—with another \$240–265 million of losses expected in the fourth quarter, and a workforce reduction of approximately 25 percent. CEO Rich Barton’s explanation bears quoting in full:

“We’ve determined the unpredictability in forecasting home prices far exceeds what we anticipated and continuing to scale Zillow Of-

fers would result in too much earnings and balance-sheet volatility.”

At the announcement Zillow owned 9,790 homes and had another 8,172 under contract to purchase—roughly 18,000 houses to liquidate. The wind-down ran through three quarters, ahead of schedule, and was complete as of September 30, 2022, when iBuying moved to discontinued operations in Zillow’s financials.

Three features of this episode matter for identification. First, the failure was a failure of *forecasting at the firm level*, not of demand in any particular ZIP code: the same announcement covered all 25 markets simultaneously, and Barton’s stated reason was the unpredictability of home prices, amplified by a company that had overridden its own algorithm’s caution. Second, the exit was *selective in exactly the direction a forecast-quality story predicts*: Opendoor—whose pricing the industry regarded as sharper and whose purchase behavior [Buchak et al. \(2025\)](#) show was disciplined by error avoidance—stayed, absorbed the same 2022 rate shock, took a \$573 million inventory write-down in the third quarter of 2022 and a \$1.353 billion net loss for fiscal 2022, and kept operating. Offerpad also stayed. RedfinNow, the smallest player, exited on November 9, 2022, a year later and for explicitly rate-driven reasons. Third, the *advisory signal survived the dealer*: the Zestimate remained on every Zillow listing after November 2, 2021. What vanished from the market was the executable quote and the deed records it generated. The exit therefore isolates the quote channel from the information-display channel—the comparison the AVM literature cannot make.

2.4 What the exit removed, by the numbers

In the ZIP codes I study, Zillow’s purchases over 2018Q2–2021Q3 averaged 0.17 percent of arm’s-length transactions, with the 90th percentile of exposed ZIPs at 0.45 percent and a 99th-percentile cap of 0.87 percent; the surviving machines (Opendoor and Offerpad combined) averaged 0.70 percent in the same ZIPs. Counting both sides of each intermediated house—the purchase deed and the resale deed—the machine’s footprint in its better markets ran to a few percent of recorded trade. These are honest numbers to keep in view: a single buyer of one house in several hundred is a meaningful marginal bidder, but it is not the market. The empirical question is what the disappearance of a buyer of *this* size—and of the public, model-generated prices its deeds

put on the record—did to the remaining market, and the model disciplines both what to expect and how big it should be.

3 A Model of Algorithmic Intermediation

The model has two layers. The first is the machine’s problem: a dealer that quotes a firm price from a noisy valuation model to sellers who know their house better than the model does. It delivers the sorting, fragility, and selective-exit results. The second layer embeds the machine’s transactions in the information set of ordinary buyers and delivers the price-discovery and liquidity spillovers. Proofs are in [Appendix A](#).

3.1 Environment

A market segment—a ZIP code, or a cell of similar houses within one—contains a continuum of houses. House i ’s value to any buyer is $v + \varepsilon_i$, where v is a segment-level common component and $\varepsilon_i \sim U[-w, w]$ is house-specific and privately observed by the seller. The half-width w indexes segment heterogeneity: it is the inverse of “machine-readability.” In a subdivision of near-identical tract homes w is small; in a neighborhood of custom houses on irregular lots it is large.

Sellers come in two types. A fraction λ are *movers*: they carry an urgency cost $D > 0$ —a job in another city, a divorce, an estate—so their net payoff from a market sale is $v + \varepsilon_i - D$, and the price they negotiate in a market sale embeds a concession θD , $\theta \in (0, 1]$, to the buyer who accommodates their timeline. The remaining fraction $1 - \lambda$ are *non-movers* with zero urgency, of whom an exogenous fraction ν list for idiosyncratic reasons and sell at fundamental value.

The machine—the algorithmic intermediary—runs an AVM that produces an estimate $\hat{v} = v + a$ of the segment’s common value, with forecast error $a \sim U[-s, s]$, so $\sigma_a^2 = s^2/3$. It posts a firm bid

$$q = \hat{v} - m \tag{1}$$

to any seller in the segment, where m is the gross spread. It pays an all-in cost c per intermediated house (transaction, holding, and renovation costs) and resells at fundamental value. The key informational asymmetry is the one

the institutions dictate: the seller observes ε_i ; the machine observes neither ε_i nor its own error a .

Assumption 1. $0 < c < D$; $2w > D - c$; $0 < s < \min\{c, (D - c)/2, 2w - (D - c)\}$.

The first condition says gains from trade exist: the machine's cost of intermediating is below the mover's urgency cost. The second says heterogeneity is not trivial relative to the liquidity gain, so the machine cannot profitably sweep the whole segment. The third bounds forecast errors by costs and by the slack in the second condition—it guarantees that no realization of the error exhausts the mover pool, keeps every threshold interior, and is the empirically relevant case: an AVM whose error rivaled the gross spread would not be deployed.

3.2 The machine's problem

A seller accepts the quote if and only if it beats the market payoff. A mover compares $q = v + a - m$ with $v + \varepsilon_i - D$ and accepts iff $\varepsilon_i \leq a - m + D$; a non-mover accepts iff $\varepsilon_i \leq a - m$. Define $\mu = m - a$, the *realized* spread relative to the truth, and the acceptance thresholds

$$\tau_\delta = a - m + \delta, \quad \delta \in \{0, D\}, \quad (2)$$

for the non-mover and mover pools respectively. Acceptance is selection from below: the machine attracts the sellers whose private draws ε_i are worst relative to its quote. The next lemma prices that selection exactly.

Lemma 1 (Pool profit identity). *For a seller pool with urgency δ and interior threshold $\tau_\delta \in [-w, w]$: the acceptance mass is $(\tau_\delta + w)/(2w)$; the mean accepted draw is $\mathbb{E}[\varepsilon \mid \text{accept}] = (\tau_\delta - w)/2$; and the pool's contribution to machine profit is*

$$\frac{(\delta - c)^2 - (w + c - \mu)^2}{4w}. \quad (3)$$

The identity packs the whole trade-off into one difference of squares. The first square is the social surplus the pool offers—urgency net of cost. The second is the wedge between the spread the machine actually realizes, μ , and the spread $w + c$ at which it would exactly break even on the marginal pool; adverse selection appears as the w inside the second square. Profit from

non-movers ($\delta = 0$) is $[c^2 - (w + c - \mu)^2]/4w$, which is negative whenever the non-mover pool is active ($\mu < w$): sellers with no urgency accept only when the machine has overpriced, so every non-mover trade is a lemon. The machine's problem is to set the spread wide enough to exclude them without choking off the movers.

Proposition 1 (Machine-readable segments). *Under Assumption 1 the machine's optimal spread is $m^* = w + c$. At the optimum only movers trade: the non-mover threshold is $\tau_0 = a - w - c < -w$ for every realization of a , since $a \leq s < c$. Expected volume is*

$$V^* = \frac{\lambda(D - c)}{2w}, \quad (4)$$

and expected profit is

$$\Pi^*(w, s) = \frac{\lambda[(D - c)^2 - s^2/3]}{4w}. \quad (5)$$

V^* and Π^* are strictly decreasing in w , and m^* is strictly increasing in w . With a per-segment fixed operating cost $F > 0$, the machine operates if and only if

$$w \leq \bar{w} = \frac{\lambda[(D - c)^2 - s^2/3]}{4F}. \quad (6)$$

Proposition 1 is the sorting result: algorithmic intermediation selects into homogeneous, machine-readable segments. The spread $m^* = w + c$ rises one-for-one with heterogeneity—the machine must charge more where it knows less—and the wider spread exactly offsets the worsening selection, leaving the per-trade margin $(D - c - a)/2$ independent of w ; volume and profit then fall in w entirely through participation, as the spread prices out more movers. With any fixed cost of standing up a market—pricing infrastructure, local renovation crews, holding facilities—the machine operates only below the heterogeneity cutoff $\bar{w}$. This is the cross-sectional prediction: iBuyer share should decline in the dispersion of the local housing stock, which is what Buchak et al. (2025) and Seiler and Yang (2023) document and what my Figure 2 shows in deeds.

Proposition 2 (Forecast-error fragility and collapse). *(i) Π^* is strictly decreasing in s^2 :*

$$\mathbb{E}_a \Pi(m^*) = \Pi(m^*; a = 0) - \frac{\lambda s^2}{12w}, \quad (7)$$

so forecast-error variance is a pure tax on the spread. The machine operates iff

$$s^2 \leq \bar{s}^2(w) = 3 \left[(D - c)^2 - \frac{4wF}{\lambda} \right], \quad (8)$$

and a regime shift that raises forecast-error dispersion above $\bar{s}$ triggers exit. (ii) (Operational winner’s curse.) Conditional on the realized error a , volume

$$V(a) = \frac{\lambda(D - c + a)}{2w} \quad (9)$$

is strictly increasing in a while the per-trade margin $(D - c - a)/2$ is strictly decreasing, so

$$\text{Cov}(V, a) = \frac{\lambda s^2}{6w} > 0 : \quad (10)$$

the algorithm buys the most precisely when it overprices, and realized profit $\Pi(a) = \lambda[(D - c)^2 - a^2]/(4w)$ is lowest in its highest-volume states. (iii) $\bar{s}^2(w)$ is decreasing in w : harder segments are more fragile.

Corollary 1 (Selective exit). *Let firms j be heterogeneous in forecast quality s_j , and suppose the scaled errors κs_j remain within Assumption 1’s bounds. A common non-stationarity shock that scales all forecast errors by $\kappa > 1$ removes exactly the operating firms with $s_j \in (\bar{s}/\kappa, \bar{s}]$: the high-error firm exits while the precise firm stays.*

Proposition 2 is the 2021 collapse as comparative statics. Part (i) says the machine’s viability is a race between the urgency surplus $(D - c)^2$ and its error variance; nothing about local demand appears in the exit condition. Part (ii) is the mechanism reporters found in the wreckage: in mid-2021, when Zillow’s model (and the human “offer calibration” layered on top of it) overpriced, conversion surged—9,680 purchases against 3,032 sales in a single quarter—exactly the high- a , high-volume, negative-margin state the covariance in (10) describes. The firm’s own disclosure—a write-down from “purchasing homes in Q3 at higher prices than current estimates of future selling prices”—is the winner’s curse stated in accounting language. And Corollary 1 explains why the shock removed Zillow and not Opendoor: when the post-pandemic market broke the stationarity that AVMs train on, the firm whose forecast error was largest crossed the exit threshold first. I can corroborate the ordering in the deeds rather than relying on Zillow’s self-published 6.9 percent off-market accuracy statistic: the within-segment dispersion of each firm’s own

purchase-price residuals—a direct empirical counterpart of σ_a —is largest for Zillow (0.188 log points) and smaller for Opendoor (0.174), the firm that survived. Selective exit by forecast quality is exactly the mechanism, and it is precisely what makes Opendoor-exposed ZIPs the right control group in Section 5.²

3.3 Price discovery: the machine’s quotes as public signals

The second layer asks what the machine’s presence does to everyone else’s information. Buyers in household-to-household trades estimate the common value v by least-squares (linear) forecasts from k recent comparable transactions. A household comp is $p_j = v + \xi_j$, with ξ_j i.i.d., mean zero, variance σ_ξ^2 ; I treat σ_ξ^2 as a primitive.³

Machine purchase prices are public deed records, and the machine’s fee structure is posted; de-spreading a machine deed by the known m^* yields the signal $\hat{v} = v + a$, with variance σ_a^2 . Crucially, this error is *common to all machine comps in the segment*: one model, one error. Buyers always harvest their k household comps, scanning back as far as needed; a machine deed among the most recent transactions arrives *on top of* the household comps, not in place of one. Buying-side information therefore depends on whether the recent record contains a machine trade. With v_m denoting the machine’s share of segment transactions, the probability that the window of the k most recent transactions contains at least one machine deed is

$$\rho(v_m) = 1 - (1 - v_m)^k, \quad (11)$$

strictly increasing and concave in v_m : even a modest machine share seeds a

²These residual dispersions also fold in my hedonic’s own error, so they overstate the pure AVM component; what the model needs is only the cross-firm *ordering*, which is robust. The realized failure was surely part governance too—the “Project Ketchup” initiative overrode the algorithm—so I read Corollary 1 as one of two reinforcing accounts of why the high-error firm exited, not as the sole one.

³In a stationary equilibrium the comp noise σ_ξ^2 is itself a fixed point: today’s buyer errors become tomorrow’s comp errors, so σ_ξ^2 solves a consistency condition involving W_B , the bargaining variance σ_b^2 , and the dispersion of ε and seller-urgency concessions. Endogenizing it should amplify the comparative statics below—machine presence lowers σ_ξ^2 , which lowers W_B further—and I conjecture it leaves every sign intact; I work with the one-step problem and do not rely on the fixed-point for any result.

large fraction of comp windows. The projection MSEs are

$$V_0 = \frac{\sigma_\xi^2}{k} \quad \text{and} \quad V_1 = \left(\frac{k}{\sigma_\xi^2} + \frac{1}{\sigma_a^2} \right)^{-1} \quad (12)$$

without and with the machine signal. Transaction prices in household trades are

$$p_i = \hat{v}_i + \varepsilon_i - \theta D \cdot \mathbf{1}\{\text{mover}\} + b_i, \quad (13)$$

where $\hat{v}_i$ is the buyer's estimate of v and $b_i \sim (0, \sigma_b^2)$ is bargaining noise. The price loads one-for-one on ε_i because a matched buyer inspects the house: inspection reveals the house-specific component—condition, layout, the lot—while what it cannot reveal is the segment's common value, which is exactly the object the comps estimate and the machine's model prices.

Proposition 3 (Price discovery). *The buyer-error component of within-segment dispersion of household-to-household prices around fundamentals is*

$$W_B(\rho) = \rho \frac{V_1^2 k}{\sigma_\xi^2} + (1 - \rho)V_0 + \rho(1 - \rho) \frac{V_1^2}{\sigma_a^2}, \quad (14)$$

where the last term is the across-group mean gap: buyers with a machine comp share the common error a , sitting at gap $= (V_1/\sigma_a^2)a$ from buyers without one. The derivative $dW_B/d\rho$ is linear in ρ and negative at both endpoints—it equals $V_1 - V_0 < 0$ at $\rho = 0$ and $V_1 - V_0 - 2V_1^2/\sigma_a^2 < 0$ at $\rho = 1$ —hence W_B is strictly decreasing in ρ , and therefore in v_m . Consequently: (i) machine presence compresses price dispersion among trades the machine is not part of; (ii) exit raises dispersion by $\rho(v_m^{\text{pre}})(V_0 - V_1 + \text{cross-terms})$, strictly increasing in the pre-exit machine share.

The economics is worth stating plainly. The machine's deeds inject a signal whose error is common rather than idiosyncratic. A buyer who sees one machine comp learns a lot ($V_1 < V_0$); but as machine comps spread, buyers increasingly share the *same* mistake a , which adds a between-group variance term $\rho(1 - \rho)V_1^2/\sigma_a^2$. The proposition shows the first force wins everywhere: dispersion falls monotonically in machine share. The common error never undoes the gain—it caps it. This is the Glosten–Milgrom intuition in a search market: the dealer's quotes do the market's information aggregation, and third parties free-ride.

A second, compositional channel works in the same direction. By Proposition 1, the machine absorbs movers with low ε_i —sellers who, in its absence, would have transacted at discounted prices in the household market. Removing them truncates the household pool from below, mechanically compressing measured dispersion. The two channels are distinguishable in data: truncation moves the tails and is invisible to a robust scale measure, while information moves the whole distribution. I therefore measure dispersion both ways—standard deviation and IQR-based—and read their divergence as the composition share of the effect.

3.4 Liquidity and the left tail

Proposition 4 (Liquidity). *(i) Recorded transaction volume without the machine is $V_{0,total} = \lambda + (1 - \lambda)\nu$, and rises one-for-one with machine purchases: an intermediated house generates two deeds—machine purchase plus machine resale—against one in the counterfactual. Exit removes this flow in proportion to pre-exit exposure. (ii) (Fire sales.) With $b = 0$ and buyers pricing at the true common value ($\hat{v}_i = v$), household trades at more than t below fundamental, for any $t \leq \theta D$, are exactly mover sales. The machine absorbs movers with $\varepsilon \leq \tau_D$, so the fire-sale share of household trades falls from*

$$\frac{\lambda}{\lambda + (1 - \lambda)\nu} \quad \text{to} \quad \frac{\lambda P(\varepsilon > \tau_D)}{\lambda P(\varepsilon > \tau_D) + (1 - \lambda)\nu}. \quad (15)$$

Exit reverses this.

The machine is a liquidity provider in the textbook sense of Duffie et al. (2005) and Weill (2007): it stands ready to buy from the agents for whom waiting is most expensive. Its presence shows up in deeds as more volume and a thinner left tail of prices; its exit shows up as the reverse, concentrated where it had been buying.

3.5 Testable predictions

The model maps to the two research designs as follows.

- **P1 (sorting; Proposition 1).** Cross-section: iBuyer share declines in ZIP-level housing-stock heterogeneity. [Figure 2; Table 4 interactions.]

- **P2 (fragility; Proposition 2, Corollary 1).** Time series: a non-stationarity shock removes the highest-error algorithm while precise rivals stay—the 2021 collapse narrative of Section 2.3. The winner’s curse (high volume in overpricing states) is Zillow’s Q3 2021.
- **P3 (price discovery; Proposition 3).** Entry event studies: within-ZIP dispersion of non-iBuyer prices falls after entry. Exit DiD: dispersion rises after November 2021 in proportion to Zillow exposure, against an Opendoor-exposure placebo; SD-based and IQR-based measures separate information from composition.
- **P4 (liquidity; Proposition 4).** Entry: volume up, left-tail share down. Exit: volume down, left-tail share up, in proportion to Zillow exposure.

4 Data

4.1 Deeds and housing characteristics

The transaction data are CoreLogic deed records (Owner Transfer), the standard near-universe of recorded U.S. property transfers, merged to CoreLogic Property Characteristics at the parcel level. I use the 22 states in which at least one iBuyer operated, from 2010Q1 through 2024Q2—a window that brackets the industry’s birth, peak, and partial death. I retain market sales: arm’s-length category-A deeds; residential single-family and condominium parcels (property indicator 10/11); prices between \$10,000 and \$10 million; excluding inter-family transfers, foreclosure and REO-related deeds. The filter waterfall is in Appendix Table A2; the final sample contains 42,382,121 arm’s-length transactions across 18,657 state-ZIP units in 525 core-based statistical areas.

4.2 Identifying the machines in deeds

iBuyers transact through dedicated subsidiaries whose names appear as the first-listed buyer or seller on the deed. I identify them with curated entity-name patterns: OPENDOOR (including the “OD Arizona/Nevada/Texas LLC” family), ZILLOW (in practice, Zillow Homes Property Trust), OFFERPAD (including the “OP SPE” family—OP SPE PHX1 LLC and sib-

lings), and REDFINNOW (RedfinNow Borrower LLC). The patterns are verified against each firm’s SEC Exhibit 21.1 subsidiary lists, so the match is built on disclosed legal entities rather than string guesswork; Appendix B reports the full pattern table. A national scan of the deed data yields 376,256 iBuyer-linked deeds: 193,054 purchases and 185,635 dispositions. Three external checks validate the match. The 2021 purchase count in my data is 67,626 against industry trackers’ estimate of roughly 70,000 nationally. The mean hedonic residual on iBuyer purchases is -2.9 log points— independently reproducing the 3.1-percentage-point purchase discount that Buchak et al. (2025) document in different data—while dispositions resell at $+0.8$ log points, the spread the business model implies. And the purchase-to-disposition pairing reproduces the institutional timeline: short holding spells, concentrated metros, the 2021 Zillow surge and the 2021Q4–2022 liquidation.

4.3 Outcomes

All outcomes are at the ZIP5-by-quarter level and built from non-iBuyer (“household”) trades; I am explicit below about where the machine’s own deeds enter.

Volume. My primary volume outcome is the log count of *household-to-household* arm’s-length transactions—the machine’s purchases and resales netted out of the cell count. This matters for the exit design: a cell’s gross transaction count mechanically contains the machine’s disappearing deeds, so part of any exit coefficient on gross volume is subtraction rather than economic contraction. Netting them out isolates the trade among parties the machine was never one of. I report the gross-volume version alongside for transparency.

Dispersion. I estimate a yearly hedonic regression per state—log price on log square footage, bedrooms, bathrooms, age, age squared, a condo indicator, and ZIP fixed effects—on non-iBuyer trades only, and take residuals. Bedrooms and bathrooms enter with missing-value indicators: assessors in some states (Arizona and Maryland among them) do not report room counts, and the hedonic must not discard whole states over one regressor. Within-cell dispersion is the standard deviation of these residuals, and, separately, the IQR divided by 1.349 (the normal-consistent robust scale). The SD responds to tails; the IQR-based measure does not; Proposition 3 says their divergence measures the composition channel. Cells require at least five household trades.

Left tail. I report two versions. The fire-sale margin of Proposition 4(ii) is the share of household trades priced below fundamentals, which I measure as the share with *raw* hedonic residual below -0.25 (roughly 22 percent below the price characteristics predict). I also report the share below -0.25 in the within-cell *demeaned* residual, which is a dispersion-tail rather than a below-predicted-price tail; the two coincide only when the cell’s mean residual is zero. The text is explicit about which is which.

Median price. The within-cell median log price, used to check that information effects are not mislabeled price-level effects.

4.4 Covariates and summary statistics

ZIP covariates come from the housing stock itself: residential parcel count, the standard deviation of log square footage (the empirical counterpart of w), the share of stock built after 1990, the single-family share, and the modal CBSA. Table 1 reports summary statistics for the estimation panels. Figure 1 plots quarterly iBuyer purchases by firm—the entry ramp, the COVID pause, the 2021 spike, and the post-shutdown step down are all visible in raw deed counts. Figure 2 plots iBuyer transaction share against ZIP-stock heterogeneity: share declines monotonically in the SD of log square footage, the sorting fact (P1) that disciplines the model.

5 Empirical Strategy

The two designs are deliberately asymmetric. Entry was chosen; exit was forced. I use the entry design to establish the joint pattern the model predicts and the exit design to identify the information externality.

5.1 Design 1: staggered entry

I define a CBSA’s entry cohort as the first quarter with at least ten iBuyer purchases—a threshold that separates operational launch from stray test purchases. The estimation panel is ZIP-by-quarter, 2012Q1–2021Q3, ending before the Zillow shutdown so the entry design is uncontaminated by the exit. Because treatment timing is staggered and effects are plausibly dynamic and heterogeneous, two-way fixed effects is unreliable (de Chaisemartin

Table 1: Sample and algorithmic-buyer activity

<i>Panel A: ZIP $\times$ quarter panel (2010Q1–2024Q2)</i>	
States	22
State-ZIP units	18,657
CBSAs	525
ZIP-quarter cells	768,414
Arm’s-length sales	42,382,121
Median sale price (\$)	173,000
Mean within-cell residual SD (log pts)	0.409
Mean within-cell residual IQR/1.349	0.310
Mean hedonic R^2 (state-year)	0.60
<i>Panel B: iBuyer purchases, 2013–2024 (national scan)</i>	
Opendoor	133,227
Offerpad	33,406
Zillow Offers	21,895
RedfinNow	4,526
Total purchases	193,054
Total dispositions	185,635
Mean purchase residual (log pts)	-0.029
Mean disposition residual (log pts)	0.008

Notes: Panel A covers the 22 iBuyer-active states: arm’s-length category-A residential (SFR + condo) deeds, \$10k–\$10M, non-interfamily, non-foreclosure/REO. Residual dispersion within ZIP-quarter cells with at least five household-to-household trades, from state-by-year hedonic regressions. Panel B: purchases identified from first-listed buyer entity names (SEC-verified aliases, Appendix Table A1).

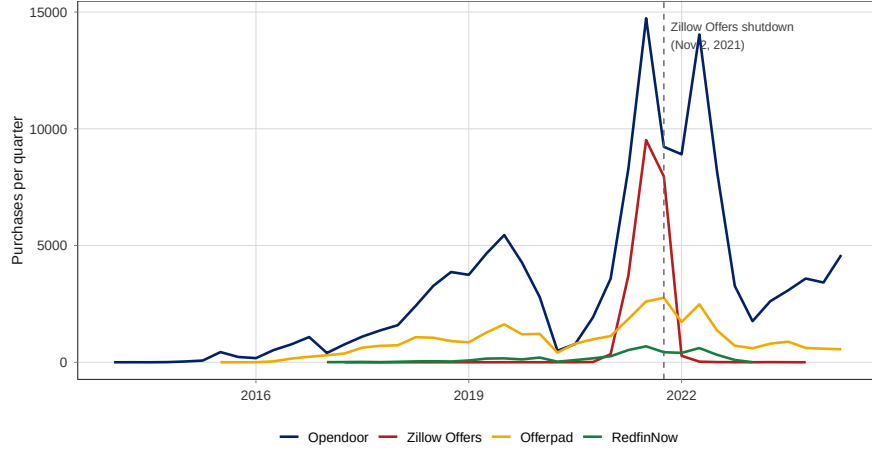


Figure 1: iBuyer purchases by firm and quarter, CoreLogic deeds, 2014–2024. Vertical lines mark the COVID pause (2020Q1) and the Zillow Offers shutdown announcement (2021Q4).

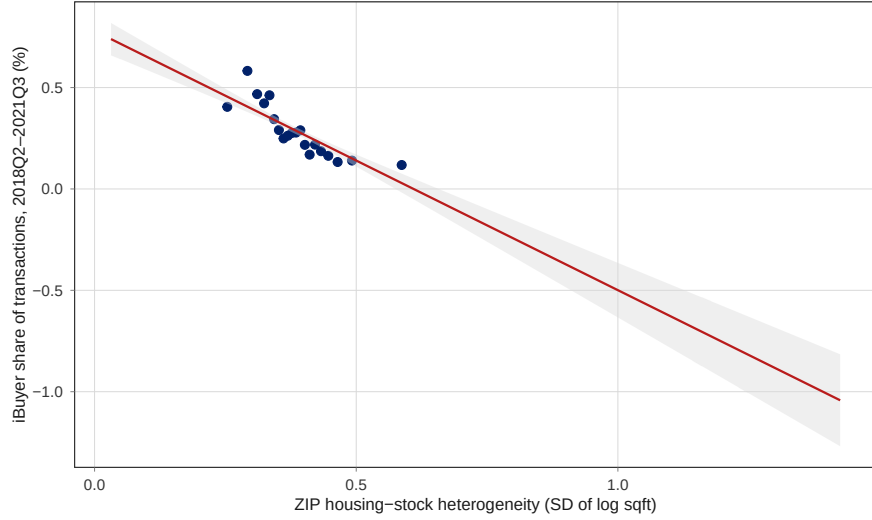


Figure 2: Machine-readability and machine presence: iBuyer share of ZIP transactions against the standard deviation of log square footage in the ZIP housing stock (binned scatter, CBSA fixed effects).

and D’Haultfœuille, 2020; Goodman-Bacon, 2021); I estimate interaction-

weighted event studies following [Sun and Abraham \(2021\)](#):

$$y_{zt} = \sum_{e \neq -1} \beta_e \mathbf{1}\{t - G_{c(z)} = e\} + \alpha_z + \delta_t + \epsilon_{zt}, \quad (16)$$

where $G_{c(z)}$ is the entry quarter of ZIP z 's CBSA, the β_e are interaction-weighted cohort averages, never-entered CBSAs serve as controls, and standard errors are clustered by CBSA. The estimator belongs to the family of heterogeneity-robust designs developed by [Callaway and Sant'Anna \(2021\)](#) and [Borusyak et al. \(2024\)](#); in robustness I add state-by-quarter fixed effects, weight by the residential stock, compare against the static two-way fixed-effects coefficient, reassign treatment one quarter early, and end the panel before COVID.

Entry is endogenous: iBuyers selected flat, liquid, homogeneous markets on observables—that is Proposition 1, and it is visible in Figure 2. The design therefore leans on timing rather than levels: the identifying assumption is that, conditional on ZIP and quarter effects, the quarter in which a metro's algorithmic buyer turned on is not correlated with counterfactual trends in second moments. Three threats deserve attention. First, selection on trends rather than levels—addressed by showing flat pre-event coefficients over a three-year window. Second, anticipation: entries were announced weeks, not years, ahead, and I probe sensitivity by reassigning treatment one quarter early. Third, COVID: the pause and restart of 2020 are absorbed by calendar-time effects common to treated and control metros, and results are robust to ending the panel in 2019Q4.

5.2 Design 2: the Zillow exit

The exit design compares ZIP codes by their exposure to the disappeared machine, within metro and quarter. Exposure is

$$\text{Zshare}_z = 100 \times \frac{\text{Zillow purchases in } z, \text{ 2018Q2–2021Q3}}{\text{total arm's-length sales in } z, \text{ 2018Q2–2021Q3}}, \quad (17)$$

winsorized at the 99th percentile. The control loading is the same object for the surviving machines: ODshare_z , the combined Opendoor-plus-Offerpad share over the same window. The sample is ZIPs with at least 50 transactions in the exposure window and at least 500 residential parcels, in CBSAs with at

least 50 Zillow purchases—markets where Zillow Offers genuinely operated—over 2019Q1–2024Q2; ZIPs in those CBSAs with zero Zillow exposure are the within-CBSA controls. The specification is

$$y_{zt} = \beta \text{Zshare}_z \times \text{Post}_t + \gamma \text{ODshare}_z \times \text{Post}_t + \alpha_z + \delta_{c(z) \times t} + e_{zt}, \quad (18)$$

with $\text{Post}_t = \mathbf{1}\{t \geq 2021\text{Q4}\}$, ZIP fixed effects, CBSA-by-quarter fixed effects, and clustering by CBSA. The event-study version replaces Post_t with quarter interactions, omitting 2021Q3.

Identification rests on the institutional argument of Section 2.3: the shutdown was driven by the failure of a national forecasting system—unpredictability that “far exceeds what we anticipated,” in the CEO’s words—not by conditions in any particular ZIP. The threat is not that Zillow chose its exit timing on local trends; it is that Zillow-exposed ZIPs differ from other ZIPs in ways that matter after 2021. The ODshare control is built for exactly that: Opendoor and Offerpad selected ZIPs on the same machine-readability criteria (Proposition 1 applies to all of them), so γ absorbs the “iBuyer-attractive ZIP” selection, and—because the surviving machines’ markets took the 2022 rate shock too—it absorbs the Sunbelt cooling that coincided with the post period. β is then identified from variation in *which* machine a ZIP was exposed to, under the assumption that, conditional on metro-quarter effects and overall iBuyer loading, Zillow-tilted ZIPs would have trended like Opendoor-tilted ZIPs absent the shutdown.

The information and liquidity outcomes meet that assumption to very different degrees, and I am explicit about which result I stake the paper on. The *information* outcomes pass their falsification tests cleanly: within-ZIP price dispersion shows no differential pre-trend (a fake-2020Q3 placebo permutation centers on zero, $p = 0.56$), so the post-shutdown null is identified. The *volume* outcome does not: exposed ZIPs—the ones Zillow bought into heavily during its 2021 ramp—were already on a differential downward volume path before the shutdown (the same placebo rejects, $p = 0.01$), and the largest declines arrive in 2023, two years late. A balance table (Appendix Table A3) shows exposure groups differ on stock composition, as the sorting model requires, but are balanced on pre-period appreciation; the surviving-machine loading is a clean placebo; and inference is robust to the 27-cluster design (wild cluster bootstrap with Webb weights, 999-draw within-CBSA permutation). But none of this repairs a pre-trend in the volume outcome itself. I therefore treat the price-discovery null as the identified contribution

and the volume decline as suggestive but confounded by Zillow’s selection into cooling markets—a distinction the data force, and one I do not paper over.

6 Results

6.1 Entry

Table 2 reports the entry estimates; Figure 3 plots the event studies. The entry design returns exactly the verdict its endogeneity warnings predict, and I report it without causal varnish. Median prices rise 1.2 log points after entry (s.e. 0.4), but on a pre-trend that rejects flatness at $p < 0.001$: the machines chased appreciating markets, and no event-study reweighting undoes that. Volume *falls* by 2.2 log points (s.e. 1.3, pre-trend $p = 0.05$)—the opposite of Proposition 4(i)’s mechanical arithmetic—which reads as the inventory tightening of the boom markets the machines selected, not as an effect of entry. The SD-based dispersion measure shows nothing (+0.15 log points, s.e. 0.27) and fails its own pre-trend test ($p = 0.01$). The single outcome whose pre-trends are clean is the robust IQR-based dispersion measure (joint $p = 0.77$): it falls by 0.39 log points after entry (s.e. 0.22, $p = 0.08$), the model-consistent sign, and the sign survives all five variants in Table 2 (estimates -0.26 to -0.63 log points). Selection visible everywhere, one clean outcome pointing the model’s way: that is what the entry data contain, and it is why identification rests on the exit. The estimation sample is 264,485 ZIP-quarters for dispersion outcomes and 320,667 for volume and prices.

6.2 Exit

Table 3 reports estimates of equation (18); Figure 4 plots the event studies. I lead with the cleanly identified result—the price-discovery null—and treat the volume decline, which is large but confounded, second.

Price discovery. When the machine’s quotes vanished, within-ZIP dispersion of household prices did not measurably rise. The SD-based estimate is +0.84 log points per percentage point of Zillow exposure (s.e. 1.06, permutation $p = 0.44$); the composition-free IQR measure—the one my own framework (Section 3.3) designates as the information channel—is -0.20 (s.e. 0.63); the raw fire-sale tail moves +0.39 percentage points (s.e. 0.67). Every

Table 2: iBuyer entry: average post-entry effects (descriptive)

Outcome	(1) Sun–Abraham	(2) State×qtr FE	(3) Stock-weighted	(4) TWFE
log volume	-0.022* (0.013)	-0.021* (0.012)	-0.031** (0.013)	-0.078** (0.037)
residual SD	0.002 (0.003)	0.000 (0.004)	0.004 (0.003)	-0.010** (0.004)
residual IQR/1.349	-0.004* (0.002)	-0.005 (0.003)	-0.003 (0.002)	-0.006 (0.004)
log median price	0.012*** (0.004)	0.005 (0.005)	0.016*** (0.004)	0.061*** (0.012)
left-tail share	0.000 (0.001)	0.000 (0.002)	0.001 (0.001)	-0.006*** (0.002)
Pre-trend joint p (vol; SD; IQR)	0.05; 0.01; 0.77			

Notes: Average post-entry effect of CBSA-level iBuyer entry (Sun and Abraham, 2021), ZIP×quarter panel 2012Q1–2021Q3, ZIP and quarter fixed effects, never-entered CBSAs as controls. The entry design is *descriptive*: pre-trends reject for all outcomes except the IQR-based dispersion measure ($p = 0.77$). SEs clustered by CBSA. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

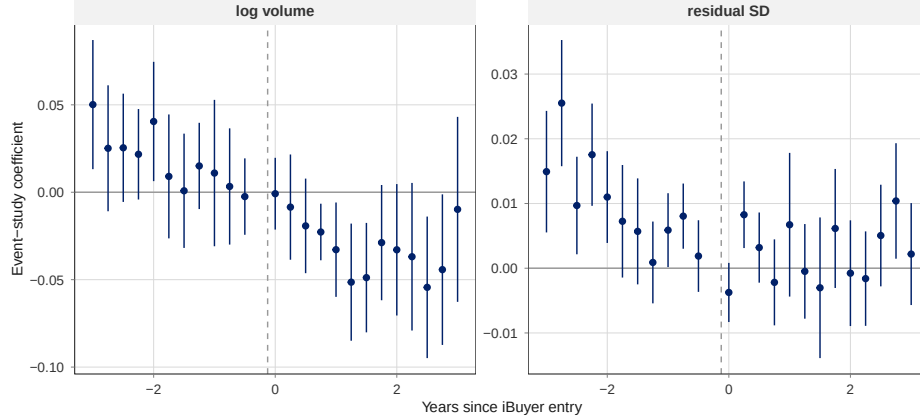


Figure 3: Entry event studies (Sun–Abraham interaction-weighted): volume, residual dispersion (SD and IQR), and left-tail share around CBSA iBuyer entry. 95% confidence intervals, CBSA-clustered.

estimate is statistically indistinguishable from zero, and the null is *identified*: the fake-2020Q3 placebo permutation centers on zero for dispersion ($p = 0.56$) and the tail ($p = 0.22$), so there is no differential pre-trend mask-

ing an effect. The bound is economically tight. At mean exposure the IQR point estimate is a fraction of a log point against a baseline of 31, and the 95 percent one-sided ceiling is 0.6 percent of baseline dispersion. Median prices in exposed ZIPs show +1.5 log points (s.e. 1.8): no differential price-level movement either. Section 7 shows this ceiling is sharp enough to reject the model’s most aggressive calibrations and to confirm its central one—a signal in under two percent of comparable-sale windows cannot move price discovery much. The dispersion sample is 49,130 ZIP-quarters; the extensive-margin check (Table 6, Panel C) finds a small, positive treatment association with cell inclusion (+0.024, $p = 0.02$)—the opposite sign from the attrition that would bias the null toward zero, so the null is not a selection artifact.

Volume. Household-to-household volume—the machine’s own deeds netted out—falls by 10.0 log points per percentage point of Zillow exposure (s.e. 3.5; permutation $p = 0.01$; wild cluster bootstrap $p = 0.01$), while the surviving-machine loading is a precise zero (−0.5 log points, s.e. 0.7); the household-only estimate is, if anything, larger than the gross-volume version (Table 3, row 2), so the decline is not the machine’s own disappearing deeds. At mean exposure that is a 1.7 percent decline; at the 90th percentile, 4.5 percent. But I do not claim this as a clean causal effect, and the event study (Figure 4) shows why. Exposed ZIPs—the ones Zillow bought into during its 2021 ramp—carry positive relative volume in 2019 (+8 to +11 log points) that drifts down toward the 2021Q3 reference, and the fake-2020Q3 placebo rejects ($p = 0.01$): these markets were already on a differential downward path before the shutdown. The post-2021Q4 coefficients step down (−7, then −13 log points in 2022Q1) but bounce back by mid-2022 and reach their largest values only in 2023—a profile more consistent with the cooling of Sunbelt markets Zillow had specifically concentrated in than with a clean break at the announcement. The dose-response (Table 6, Panel B) is significant at every exposure tercile and roughly flat (−7.3, −6.5, −6.4 log points), which rules out a handful of extreme-dose ZIPs driving the slope but also shows the response is closer to an extensive “any-Zillow” margin than a linear dose. And the vintage split (Panel A) is uninformative for the cleanest test: Zillow’s pre-2021 purchasing in these ZIPs was negligible (the pre-Ketchup exposure mean is essentially zero), so almost all the identifying variation is the 2021 ramp itself—exactly the froth-era buying whose reversion I cannot rule out. The honest conclusion is that the volume decline is large and robust to the wind-down (the donut estimate is −8.7, s.e. 3.6) but not cleanly separable from Zillow’s selection into cooling markets.

A second, smaller event sharpens the caution rather than dispelling it. RedfinNow—a minor iBuyer that never ramped in 2021—shut down in November 2022; its exit shows no volume effect (+0.1 log points per point of exposure, s.e. 2.8, in 14 CBSAs). The liquidity-multiplier reading predicts a scaled-down replication and does not get one; whether that reflects RedfinNow’s tiny footprint (mean exposure under a tenth of a percent, leaving the test underpowered) or the absence of a genuine causal channel, the data cannot say. Taken together, the exit evidence *identifies* a tight null on price discovery and *documents* a large but confounded volume decline.

Table 3: The Zillow Offers shutdown: difference-in-differences

Outcome	Zillow share × Post	Opendoor/Offerpad share × Post	N
Log volume (household)	-0.1004*** (0.0350)	-0.0047 (0.0068)	53,343
Log volume (incl. machine deeds)	-0.0804** (0.0354)	-0.0066 (0.0069)	53,343
Residual SD	0.0084 (0.0106)	-0.0058 (0.0039)	49,130
Residual IQR/1.349	-0.0020 (0.0063)	-0.0029 (0.0023)	49,130
Left-tail share (demeaned)	0.0052 (0.0054)	-0.0021 (0.0028)	49,130
Left-tail share (raw)	0.0039 (0.0067)	-0.0045 (0.0039)	49,130
Log median price	0.0155 (0.0176)	0.0037 (0.0047)	53,343

Notes: ZIP×quarter panel, 2019Q1–2024Q2, ZIPs in CBSAs with ≥ 50 Zillow Offers purchases (2425 ZIPs, 27 CBSAs, 940 zero-exposure controls). The *primary* volume outcome is household-only (machine purchases and resales netted out); the gross row is shown for transparency. Exposure = firm purchases as a percent of ZIP transactions over 2018Q2–2021Q3, winsorized at the 99th percentile. Post = from 2021Q4 (announced Nov 2, 2021). ZIP and CBSA×quarter fixed effects; SEs clustered by CBSA (27 clusters). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

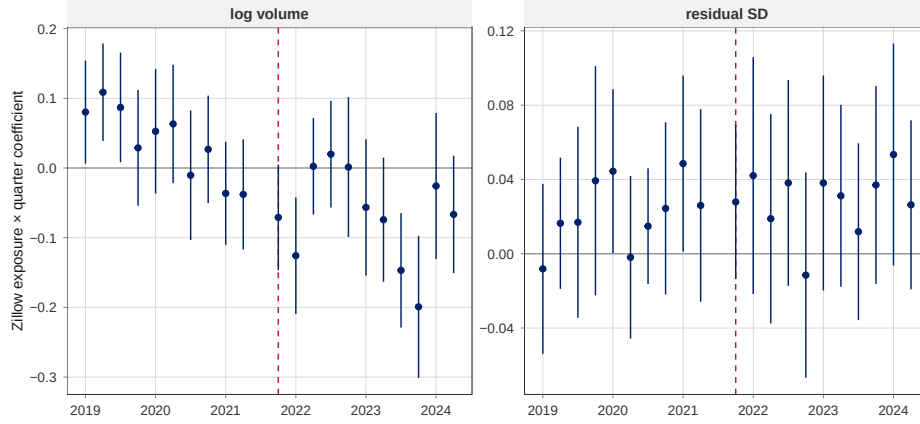


Figure 4: Exit event study: quarterly coefficients on household-volume and dispersion $Zshare_z$ interactions (reference 2021Q3), through 2024Q2, with ODshare interactions as the placebo loading. 95% confidence intervals, CBSA-clustered. The vertical line marks the November 2, 2021 shutdown announcement.

6.3 Heterogeneity by machine-readability

Proposition 3 ties the size of the information spillover to the machine’s local share, and Proposition 1 ties that share to stock homogeneity. The sorting itself is sharp in the data: Figure 2 shows the iBuyer transaction share falling by 1.28 percentage points per unit of stock-heterogeneity (the SD of log square footage) across 8,509 ZIPs—the machines went where the machines could read. But sorting (a cross-section in the share the machine *achieves*) and spillover heterogeneity (how the exit *effect* varies across segments) are different objects, and the second is not resolvable at these effect sizes. Table 4 splits the designs by readability tercile; the estimates are uniformly insignificant. With the average spillover bounded near zero, its cross-segment pattern is below the design’s power, and I report the splits for completeness rather than as a test of Proposition 3.

6.4 Robustness

Table 5 collects the full robustness grid for the two primary outcomes, with the Opendoor/Offerpad placebo loading near zero throughout. The *magnitude* of the household-volume estimate is stable—dropping the liquidation

Table 4: Heterogeneity by machine readability (dispersion outcome)

ZIP readability tercile	Exit: Zillow share $\times$ Post	Entry: Post
Least machine-readable (T1)	0.0080 (0.0175)	(ref.)
Middle (T2)	0.0163 (0.0124)	-0.0009 (0.0032)
Most machine-readable (T3)	0.0013 (0.0102)	-0.0022 (0.0039)

Notes: Readability terciles (on distinct ZIPs) from the (negative) SD of log square footage of the ZIP residential stock; T3 = most homogeneous. Outcome: within-cell residual SD. Entry column reports effects relative to T1 (the level effect is absorbed by CBSA $\times$ quarter FE). All point estimates insignificant; reported for completeness. SEs clustered by CBSA.

quarters (-10.2 log points), ZIP-level clustering (-10.0), transaction weighting (-9.3), 95th-percentile winsorization (-9.8), a ten-trade cell minimum (-11.1), and excluding Phoenix (-9.5) all leave it between roughly -9 and -11 , with permutation and wild-bootstrap p -values below 0.05; only the top-quartile binary coarsens it (-2.2), consistent with the flat tercile dose-response. I stress again that this stability is stability of a *magnitude* whose causal interpretation the pre-trend clouds, not a robustness that resolves the confound. The price-discovery bound is stable in the sense that matters for it: every cell of the grid leaves the dispersion effect between 0.1 and 1.6 log points and statistically indistinguishable from zero. For the descriptive entry design, the IQR-dispersion estimate is -0.31 log points reassigning treatment a quarter early and -0.63 (s.e. 0.20) ending the panel in 2019Q4 to excise COVID; the columns of Table 2 bracket the baseline.

7 Interpretation

7.1 Magnitudes: the information value of a firm quote

Start with the result the design identifies: the price-discovery bound. Read off the composition-free IQR measure, the exit effect is -0.20 log points per exposure point and statistically zero; the SD measure is $+0.84$ and equally insignificant; both have clean pre-period placebos. Rather than dollarize a

Table 5: Exit design: robustness and inference

Outcome	(1) Drop liq.	(2) Top- quart.	(3) ZIP clust.	(4) Weight	(5) Wins. 95th	(6) Excl. Phx	(7) Perm. p	(8) Wild p
Log volume (household)	-0.1015*** (0.0363)	-0.0222* (0.0127)	-0.1004*** (0.0279)	-0.0925*** (0.0295)	-0.0979*** (0.0344)	-0.0946** (0.0348)	0.010	0.008
Residual SD	0.0086 (0.0103)	0.0009 (0.0043)	0.0084 (0.0077)	0.0159 (0.0125)	0.0073 (0.0119)	0.0072 (0.0120)	0.442	0.510

Notes: Zillow-share $\times$ Post coefficient under variants of Table 3 (primary household-volume and SD-dispersion outcomes). (1) drops 2021Q4–2022Q2; (2) top-quartile exposure indicator; (3) ZIP clustering; (4) transaction-weighted; (5) 95th-pct winsorization; (6) excluding Phoenix; (7) permutation p from 999 within-CBSA reshuffles, $(1 + \#\{|t| \geq |t_{obs}|\})/(B + 1)$; (8) wild cluster bootstrap (Webb weights, 999 reps, null imposed). The Opendoor/Offerpad placebo loading is near zero in every variant (Appendix).

Table 6: Identification checks (household volume)

	Coefficient	(s.e.)
<i>Panel A: vintage split of exposure</i>		
Pre-Ketchup exposure (2018Q2–2020Q4) $\times$ Post	2.0623	(1.5658)
Ramp exposure (2021Q1–Q3) $\times$ Post	-0.1070***	(0.0356)
<i>Panel B: dose-response (exposure terciles vs. zero)</i>		
Tercile 1 $\times$ Post	-0.0734***	(0.0188)
Tercile 2 $\times$ Post	-0.0650***	(0.0185)
Tercile 3 $\times$ Post	-0.0636***	(0.0195)
<i>Panel C: extensive margin (cell has ≥ 5 trades)</i>		
Zillow share $\times$ Post	0.0245**	(0.0097)

Notes: All specifications carry ZIP and CBSA $\times$ quarter fixed effects, the Opendoor/Offerpad control loading, and CBSA-clustered SEs. Panel A splits Zillow exposure by purchase vintage. Panel B replaces the continuous dose with exposure-tercile indicators (distinct ZIPs; zero- exposure ZIPs are the omitted group). Panel C regresses an indicator for the dispersion cell clearing the five-trade minimum on the treatment, to test treatment-correlated sample selection.

point estimate indistinguishable from zero, I state a ceiling: the data rule out an information externality larger than about 0.6 percent of baseline dispersion at the achieved scale (Table A4). What makes this bound economically meaningful is that the model, fed its inputs from data rather than assumptions, predicts an effect in the same small range. Zillow’s deed flow at mean exposure is $v_m \approx 0.3$ percent of segment transactions; with comp windows of $k \in \{4, 6, 8\}$, equation (11) gives the quote a reach of ρ around one to two percent of buyers. The signal’s error σ_a I take not from a free parameter but from Zillow’s *own* within-segment purchase-price residual dispersion in my deeds, 0.188 in log units—above the roughly ten-percent standard deviation implied by Zillow’s published 6.9 percent median off-market error, because it also absorbs my hedonic’s own error, so it is a conservative (large) input. Feeding these into Proposition 3, the predicted dispersion effect of the exit ranges up to 0.45 log points across the parameter grid (Table A4)—and here the bound has teeth: the data’s IQR ceiling, 0.17 log points at mean exposure, lies *below* the model’s most aggressive cells (large comp windows and a high buyer-error share), so the design actually rejects them, while it is consistent with the model’s central calibrations. The honest reading is therefore neither “confirmed” nor “no power”: the externality is real in the model and bounded tightly in the data, and the data are sharp enough to say it was, at most, a fraction of a percent of baseline pricing dispersion. A public signal reaching one comparable-sale window in fifty cannot move segment-level price discovery much, and it did not.

The volume decline is the larger number but the weaker claim, and I treat it accordingly. At mean exposure the exit is associated with a 1.7 percent fall in household trade, and the estimate survives netting out the machine’s own deeds, dropping the wind-down, and the full robustness grid. Were it clean, the natural reading would be that the firm quote was an outside option that made marginal listings happen—a homeowner with a guaranteed cash exit lists and negotiates differently than one without—amplified by the thin-market externality of Ngai and Tenreiro (2014); and the model would organize the sign and incidence while leaving the multiplier (Proposition 4(i) predicts only the machine’s mechanical flow) to future structural work on the listing margin. But the event study and placebo say the estimate is not clean: exposed ZIPs were cooling relative to controls before the shutdown, and the largest declines come two years later, so an unknown share of the 1.7 percent is the reversion of the 2021 boom Zillow had chased rather than the loss of its standing bid. I therefore offer the liquidity channel as a hypoth-

esis the data are consistent with, not a result the design establishes—and note that distinguishing the two would take an instrument for exposure that is orthogonal to local price trajectories, which the iBuyer setting does not naturally provide.

The comparison with the equities microstructure literature is then instructive precisely in the negative. [Hendershott et al. \(2011\)](#) found that algorithmic trading narrowed spreads and made quotes more informative; [Brogaard et al. \(2014\)](#) found that high-frequency traders trade in the direction of permanent price changes. One might have expected the first market-making capital housing ever had to do the same for price discovery. It did not—not because algorithmic quotes are uninformative, but because of scale: in a limit-order book the algorithms *are* the price, while in housing the machine wrote one deed in several hundred, reached one comp window in fifty, and its removal left the price system statistically undisturbed. The open question the episode poses is what the externality would be at ten times the share, where $\rho(v_m)$ approaches one-fifth of comp windows and, by the same calibration, the model’s predicted effect rises into the range the data could detect. That is the experiment a future at-scale deployment would run; the 2021 episode ran a smaller one, and its answer for price discovery is a well-measured almost-nothing.

7.2 Welfare and fragility

The welfare reading must respect the asymmetry in what the data establish. The cleanest entry is a *non*-finding with a tight bound: at the scale the industry reached, the machine’s quotes produced no measurable price-discovery externality, positive or negative, for the households around it—the information commons one might have feared losing was, on the evidence, never large. That is reassuring for the bystanders and sobering for the technology’s boosters in equal measure. The second entry, liquidity, is where the welfare stakes plausibly lie—a standing algorithmic bid is exactly the outside option thin housing markets lack—but it is also where my identification is weakest, so I can document a large volume decline without certifying its surplus consequences; pinning them down is the first-order task for future work with cleaner exposure variation. The third entry, fragility, the model delivers even though the experiment cannot fully adjudicate it: [Proposition 2](#) says an institution that provides a market’s marginal liquidity (and, at larger scale, its price signal) is one regime shift from removing both, and [Corollary 1](#)—

corroborated by Zillow’s larger deed-level forecast-error dispersion—says the removal comes precisely when forecasting is hardest. A market that leans on one private algorithm inherits that algorithm’s failure modes. This is the [Grossman and Stiglitz \(1980\)](#) tension with a balance sheet attached, and it is the durable lesson of the episode regardless of how the liquidity magnitude is eventually resolved.

The concentration point deserves emphasis, with a caveat about scale. Price discovery in equities is produced by thousands of competing algorithms; the housing market’s brief experiment ran on three or four, of which one—the one with the larger forecast error—was the second-largest. The instinct that this concentration made the market fragile to the algorithm’s failure is right about the mechanism but, on this evidence, wrong about the magnitude for *price discovery*: the machine was simply too small a share of the comparable-sale record for its exit to register. The concentration that matters is prospective. If algorithmic market-making returns at ten times the scale, the same calibration that rationalizes today’s null implies a price-discovery externality the market would feel, produced by a handful of private models trained on a common stationarity—and removed together when that stationarity breaks.

7.3 Policy

Two regulatory margins follow directly. First, AVM oversight: bank regulators have moved toward quality-control standards for automated valuation models used in lending. The model enlarges the stakes—AVM errors need not stay inside the firm that makes them. When a model’s output is an executable quote whose deed becomes a public comp, forecast error propagates into the information environment of every neighboring transaction, a feedback the Zestimate literature has documented in advisory form ([Fuet al., 2022](#)) and that the firm-quote channel amplifies with capital. Second, appraisal waivers: the GSEs increasingly waive in-person appraisals in favor of model-based collateral assessment in exactly the homogeneous, machine-readable segments where iBuyers operated (Proposition 1 describes both selections). The exit experiment is a caution about correlated failure: if both the market’s marginal buyer and its collateral assessor are statistical models trained on the same stationarity, the states of the world in which one fails are the states in which both do.

8 Conclusion

For roughly seven years, the largest decentralized asset market in the world ran an experiment it did not know it was running. Statistical models, backed by corporate balance sheets, posted firm prices on hundreds of thousands of American homes—the first market-making capital that housing has ever had. This paper asked what that did to price discovery for everyone else, and used the abrupt, forecasting-driven death of the second-largest machine to find out.

The answer, stated honestly, is mostly a well-measured null. When the machine’s quotes vanished, the price discovery of the households around it did not measurably suffer: within-ZIP dispersion held steady, the estimate is statistically zero with a clean pre-period placebo, and the bound is tight enough to reject the larger effects one might have feared. A model of algorithmic intermediation explains why—the machine reached one comparable-sale window in fifty, too small a share of the record to move segment-level price discovery—and the same model fed Zillow’s own deed-level forecast errors predicts an effect inside the bound. Household transaction volume *did* fall sharply in exposed markets, hinting that the machine’s standing bid had supported trade among third parties; but those markets were cooling before the shutdown, so I document that decline without claiming it cleanly, and leave its causal magnitude to a design with exposure variation orthogonal to local price trends. The theory carries more than the reduced form does: it organizes where the machines operated (homogeneous, machine-readable segments, visibly so in the deeds), and why the high-forecast-error firm was the one that died (the operational winner’s curse that Zillow’s final quarter exhibited almost line by line, with its larger deed-level error confirmed against the surviving rival’s). The Zestimate, the advisory signal, never blinked throughout.

The broader lesson is about the industrial organization of information in decentralized markets, and it is more measured than the boosters or the critics of AI in housing would have it. A statistical dealer with real capital at risk is a genuinely new institution, and at the scale this one reached it left the price system almost exactly as it found it: no measurable sharpening, no measurable degradation, a public-good externality bounded to near zero. That is itself a finding—the first deployment of algorithmic market-making in the largest consumer asset market did not, on the evidence, change how well that market discovered prices. What the model insists, and what the

episode cannot refute, is that this verdict is a function of *scale*: at ten times the share the same arithmetic delivers an externality the market would feel, produced by a few private models trained on a common stationarity and apt to fail together when it breaks. The 2021 collapse ran the small version of that experiment and returned a quiet answer. The next deployment may run the large one, and it may not announce itself as politely.

References

- Akerlof, George A.**, “The Market for “Lemons”: Quality Uncertainty and the Market Mechanism,” *Quarterly Journal of Economics*, 1970, 84 (3), 488–500.
- Anenberg, Elliot**, “Information Frictions and Housing Market Dynamics,” *International Economic Review*, 2016, 57 (4), 1449–1479.
- Assad, Stephanie, Robert Clark, Daniel Ershov, and Lei Xu**, “Algorithmic Pricing and Competition: Empirical Evidence from the German Retail Gasoline Market,” *Journal of Political Economy*, 2024, 132 (3), 723–771.
- Biais, Bruno, Thierry Foucault, and Sophie Moinas**, “Equilibrium Fast Trading,” *Journal of Financial Economics*, 2015, 116 (2), 292–313.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess**, “Revisiting Event-Study Designs: Robust and Efficient Estimation,” *Review of Economic Studies*, 2024, 91 (6), 3253–3285.
- Brogaard, Jonathan, Terrence Hendershott, and Ryan Riordan**, “High-Frequency Trading and Price Discovery,” *Review of Financial Studies*, 2014, 27 (8), 2267–2306.
- Buchak, Greg, Gregor Matvos, Tomasz Piskorski, and Amit Seru**, “Why Is Intermediating Houses So Difficult? Evidence from iBuyers,” Working Paper 28252, National Bureau of Economic Research 2025. December 2020, revised June 2025. First draft November 2019; previously circulated as “iBuyers: Liquidity in Real Estate Markets?” Reported under revision at the *Journal of Political Economy* (unverified; cite as NBER WP until published).

- Calder-Wang, Sophie and Gi Heung Kim**, “Algorithmic Pricing in Multifamily Rentals: Efficiency Gains or Price Coordination?,” 2026. Working paper, SSRN No. 4403058; version February 2026. Also circulated as “Coordinated vs Efficient Prices: The Impact of Algorithmic Pricing on Multifamily Rental Markets.” VERIFY journal status.
- Callaway, Brantly and Pedro H. C. Sant’Anna**, “Difference-in-Differences with Multiple Time Periods,” *Journal of Econometrics*, 2021, 225 (2), 200–230.
- Calvano, Emilio, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello**, “Artificial Intelligence, Algorithmic Pricing, and Collusion,” *American Economic Review*, 2020, 110 (10), 3267–3297.
- Case, Karl E. and Robert J. Shiller**, “The Efficiency of the Market for Single-Family Homes,” *American Economic Review*, 1989, 79 (1), 125–137.
- de Chaisemartin, Clément and Xavier D’Haultfœuille**, “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects,” *American Economic Review*, 2020, 110 (9), 2964–2996.
- Duffie, Darrell, Nicolae Gârleanu, and Lasse Heje Pedersen**, “Over-the-Counter Markets,” *Econometrica*, 2005, 73 (6), 1815–1847.
- Fu, Runshan, Ginger Zhe Jin, and Meng Liu**, “Does Human-Algorithm Feedback Loop Lead to Error Propagation? Evidence from Zillow’s Zestimate,” Working Paper 29880, National Bureau of Economic Research 2022. Previously circulated as “Human-Algorithm Interactions: Evidence from Zillow.com.” Working paper as of mid-2025; VERIFY journal status before final submission.
- , **Yan Huang, Nitin Mehta, Param Vir Singh, and Kannan Srinivasan**, “Unequal Impact of Zestimate on the Housing Market,” *Marketing Science*, 2025, 44 (6), 1407–1427.
- Fuster, Andreas, Paul Goldsmith-Pinkham, Tarun Ramadorai, and Ansgar Walther**, “Predictably Unequal? The Effects of Machine Learning on Credit Markets,” *Journal of Finance*, 2022, 77 (1), 5–47.
- Genesove, David and Lu Han**, “Search and Matching in the Housing Market,” *Journal of Urban Economics*, 2012, 72 (1), 31–45.

- Giacoletti, Marco**, “Idiosyncratic Risk in Housing Markets,” *Review of Financial Studies*, 2021, *34* (8), 3695–3741.
- Glaeser, Edward L. and Charles G. Nathanson**, “An Extrapolative Model of House Price Dynamics,” *Journal of Financial Economics*, 2017, *126* (1), 147–170.
- Glosten, Lawrence R. and Paul R. Milgrom**, “Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders,” *Journal of Financial Economics*, 1985, *14* (1), 71–100.
- Goodman-Bacon, Andrew**, “Difference-in-Differences with Variation in Treatment Timing,” *Journal of Econometrics*, 2021, *225* (2), 254–277.
- Grossman, Sanford J. and Joseph E. Stiglitz**, “On the Impossibility of Informationally Efficient Markets,” *American Economic Review*, 1980, *70* (3), 393–408.
- Guren, Adam M.**, “House Price Momentum and Strategic Complementarity,” *Journal of Political Economy*, 2018, *126* (3), 1172–1218.
- Han, Lu and William C. Strange**, “The Microstructure of Housing Markets: Search, Bargaining, and Brokerage,” in Gilles Duranton, J. Vernon Henderson, and William C. Strange, eds., *Handbook of Regional and Urban Economics*, Vol. 5, Elsevier, 2015, pp. 813–886.
- Harrison, David M., Michael J. Seiler, and Liuming Yang**, “The Impact of iBuyers on Housing Market Dynamics,” *Journal of Real Estate Finance and Economics*, 2024, *68* (3), 425–461.
- Hendershott, Terrence, Charles M. Jones, and Albert J. Menkveld**, “Does Algorithmic Trading Improve Liquidity?,” *Journal of Finance*, 2011, *66* (1), 1–33.
- Jiang, Erica Xuwei, Nadia Kotova, and Anthony Lee Zhang**, “Liquidity in Residential Real Estate Markets,” 2024. Working paper, SSRN No. 3386353, draft March 2024. VERIFY journal status before final submission.
- Krainer, John**, “A Theory of Liquidity in Residential Real Estate Markets,” *Journal of Urban Economics*, 2001, *49* (1), 32–53.

- Kurlat, Pablo and Johannes Stroebel**, “Testing for Information Asymmetries in Real Estate Markets,” *Review of Financial Studies*, 2015, *28* (8), 2429–2461.
- Kyle, Albert S.**, “Continuous Auctions and Insider Trading,” *Econometrica*, 1985, *53* (6), 1315–1335.
- Ngai, L. Rachel and Silvana Tenreyro**, “Hot and Cold Seasons in the Housing Market,” *American Economic Review*, 2014, *104* (12), 3991–4026.
- Piazzesi, Monika, Martin Schneider, and Johannes Stroebel**, “Segmented Housing Search,” *American Economic Review*, 2020, *110* (3), 720–759.
- Seiler, Michael J. and Liuming Yang**, “The Burgeoning Role of iBuyers in the Housing Market,” *Real Estate Economics*, 2023, *51* (3), 721–753.
- Stroebel, Johannes**, “Asymmetric Information about Collateral Values,” *Journal of Finance*, 2016, *71* (3), 1071–1112.
- Sun, Liyang and Sarah Abraham**, “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects,” *Journal of Econometrics*, 2021, *225* (2), 175–199.
- Weill, Pierre-Olivier**, “Leaning Against the Wind,” *Review of Economic Studies*, 2007, *74* (4), 1329–1354.
- Wheaton, William C.**, “Vacancy, Search, and Prices in a Housing Market Matching Model,” *Journal of Political Economy*, 1990, *98* (6), 1270–1292.
- Yu, Shuyi**, “Algorithmic Outputs as Information Source: The Effects of Zestimates on Home Prices and Racial Bias in the Housing Market,” 2020. Working paper, SSRN No. 3584896. VERIFY year of latest draft and journal status.

Appendix

A Proofs

Throughout, total machine profit at realized error a is the mass-weighted sum of pool contributions, λ times the mover-pool contribution plus $(1 - \lambda)$ times the non-mover-pool contribution, each given by Lemma 1 when the pool's threshold is interior. Recall $\mu = m - a$, $\tau_D = D - \mu$, $\tau_0 = -\mu$.

A.1 Proof of Lemma 1

A seller in the pool with urgency δ accepts iff $\varepsilon_i \leq \tau_\delta = a - m + \delta$. With $\varepsilon_i \sim U[-w, w]$ and $\tau_\delta \in [-w, w]$, the acceptance mass is $(\tau_\delta + w)/(2w)$ and the accepted draws are uniform on $[-w, \tau_\delta]$, so $\mathbb{E}[\varepsilon \mid \text{accept}] = (\tau_\delta - w)/2$. The machine pays $q = v + a - m$, incurs c , and resells house i at its fundamental value $v + \varepsilon_i$; per-trade profit is therefore $(v + \varepsilon_i) - (v + a - m) - c = m - a - c + \varepsilon_i$, with conditional mean $m - c - a + (\tau_\delta - w)/2$. Write $A = \tau_\delta + w$ and $B = 2(m - a - c) + \tau_\delta - w$, so the pool contribution is

$$\frac{A}{2w} \cdot \frac{B}{2} = \frac{AB}{4w} = \frac{(A + B)^2 - (A - B)^2}{16w}.$$

Direct computation gives $A + B = 2\tau_\delta + 2(m - a - c) = 2(\delta - c)$ and $A - B = 2w - 2(m - a - c) = 2(w + c - \mu)$. Hence the contribution equals $[(\delta - c)^2 - (w + c - \mu)^2]/(4w)$. ■

A.2 Proof of Proposition 1

Step 1: the deterministic problem. Fix the realized spread μ and consider profit $\Pi(\mu)$ as a function of μ .

(a) *Both-pools branch*, $\mu \in [D - w, w]$. Here $\tau_D = D - \mu \in [D - w, w] \subseteq [-w, w]$ and $\tau_0 = -\mu \in [-w, w]$, so both pools are interior and, by Lemma 1,

$$\Pi(\mu) = \frac{\lambda(D - c)^2 + (1 - \lambda)c^2 - (w + c - \mu)^2}{4w},$$

which is strictly increasing in μ on this branch because $\mu \leq w < w + c$.

(b) *Movers-only branch*, $\mu \in [w, D + w]$. Here $\tau_0 = -\mu \leq -w$: the non-mover pool is inactive. The mover pool gives

$$\Pi(\mu) = \frac{\lambda[(D - c)^2 - (w + c - \mu)^2]}{4w},$$

strictly increasing for $\mu < w + c$, strictly decreasing for $\mu > w + c$, with unique peak at $\mu = w + c$, which lies in the branch interior since $c > 0$ and $w + c < D + w$. (If $D > 2w$, the sub-region $\mu \in [w, D - w)$ has $\tau_D > w$, all movers accept, and $\Pi(\mu) = \lambda(\mu - c)$, strictly increasing; profit still rises monotonically into the interior region, so the peak is unaffected.)

Continuity at $\mu = w$. Branch (a) at $\mu = w$: $(w + c - w)^2 = c^2$, giving $[\lambda(D - c)^2 + (1 - \lambda)c^2 - c^2]/4w = \lambda[(D - c)^2 - c^2]/(4w)$. Branch (b) at $\mu = w$ gives the same value. The deterministic profit is therefore continuous, strictly increasing up to $\mu = w + c$, and strictly decreasing beyond: the unique maximizer is $\mu = w + c$.

Step 2: the stochastic problem. The machine fixes m ex ante, before a realizes; $\mu = m - a$ is then random on $[m - s, m + s]$.

(c) *Interior range.* Let $\underline{m} = \max\{w, D - w\} + s$. For $m \in [\underline{m}, D + w - s]$, every realization satisfies $\mu \in [\max\{w, D - w\}, D + w]$: non-movers are excluded ($\mu \geq w$) and the mover pool is never exhausted ($\mu \geq D - w$, so $\tau_D \leq w$), hence the movers-only quadratic branch applies state by state, and

$$\begin{aligned}\mathbb{E}_a \Pi(m) &= \frac{\lambda}{4w} \mathbb{E}_a [(D - c)^2 - (w + c - m + a)^2] \\ &= \frac{\lambda [(D - c)^2 - (w + c - m)^2 - s^2/3]}{4w},\end{aligned}$$

using $\mathbb{E}[a] = 0$ and $\mathbb{E}[a^2] = s^2/3$. This is uniquely maximized at $m^* = w + c$, which is interior to the range: $w + s < w + c$ because $s < c$; $D - w + s < w + c$ because $s < 2w - (D - c)$ (the third clause of Assumption 1); and $w + c \leq D + w - s$ because $s \leq (D - c)/2 < D - c$.

(d) *Lower corner, $m < \underline{m}$.* Realizations with $\mu < w$ activate the non-mover pool, whose contribution $(1 - \lambda)[c^2 - (w + c - \mu)^2]/(4w)$ is strictly negative there (for $\mu < w$, $w + c - \mu > c$); non-mover contributions are never positive. Realizations with $\mu < D - w$ (possible when $D > 2w$) put the whole mover pool into acceptance, where the mover contribution is $\lambda(\mu - c)$. Dropping the non-movers and extending the mover-pool quadratic bounds profit from above. Two cases make the extension rigorous. If $m < 2c - D - w + s$, then every realization has $\mu < 2c - D - w + 2s < c$ (using $2s < D - c$), and since $\mathbb{E}[\varepsilon \mid \text{accept}] \leq 0$ every accepted trade carries expected margin at most $\mu - c < 0$: profit is nonpositive and the claim is immediate. Otherwise every realization has $\mu \geq 2c - D - w$, and the mover contribution is bounded by the interior-branch expression pointwise: on the all-accept sub-region $\mu \leq D - w$ the gap $h(\mu) = (D - c)^2 - (w + c - \mu)^2 - 4w(\mu - c)$ is

concave in μ with roots at $\mu = 2c - D - w$ and $\mu = D - w$, hence nonnegative between them. Therefore

$$\begin{aligned}\Pi(m) &\leq \frac{\lambda}{4w} \mathbb{E}_a[(D - c)^2 - (w + c - m + a)^2] \\ &= \frac{\lambda[(D - c)^2 - (w + c - m)^2 - s^2/3]}{4w} < \mathbb{E}_a\Pi(m^*),\end{aligned}$$

where the final inequality is strict because $m < \underline{m} < w + c$ implies $(w + c - m)^2 > 0$.

(e) *Upper corner*, $m > D + w - s$. Here $\mu = m - a > D + w - 2s \geq w + c$ (using $D - c \geq 2s$), so only movers can trade and trade occurs only when $\tau_D = D - \mu \geq -w$, i.e., $\mu \leq D + w$; otherwise realized profit is zero. Realized profit is $\lambda \max\{0, [(D - c)^2 - (w + c - \mu)^2]/(4w)\}$, and its expectation is decreasing in m on this corner (each realization's $|w + c - \mu|$ grows with m where profit is positive). Evaluating at the corner's best point, $w + c - m = -(D - c - s)$, every realization is interior and

$$\begin{aligned}\mathbb{E}_a\Pi(m) &\leq \frac{\lambda[(D - c)^2 - (D - c - s)^2 - s^2/3]}{4w} \\ &\leq \frac{\lambda[(D - c)^2 - (D - c - s)^2]}{4w} \leq \mathbb{E}_a\Pi(m^*),\end{aligned}$$

where the last inequality holds because $(D - c - s)^2 \geq s^2/3$, which follows from $D - c \geq 2s \geq s(1 + 1/\sqrt{3})$.

Step 3: the optimum. Steps (c)–(e) establish that $m^* = w + c$ is the global maximizer. At m^* : the non-mover threshold is $\tau_0 = a - w - c < -w$ for every realization since $a \leq s < c$, so only movers trade. Realized volume is $\lambda(\tau_D + w)/(2w) = \lambda(D - c + a)/(2w)$, so expected volume is $V^* = \lambda(D - c)/(2w)$, and expected profit is $\Pi^*(w, s) = \lambda[(D - c)^2 - s^2/3]/(4w)$ from (c). V^* and Π^* are strictly decreasing in w (both scale as $1/w$ with positive numerators, the latter positive because $s^2 < (D - c)^2/4 < (D - c)^2 \cdot 3$), and $m^* = w + c$ is strictly increasing in w . With fixed cost $F > 0$ the machine operates iff $\Pi^*(w, s) \geq F$, i.e., iff $w \leq \bar{w} = \lambda[(D - c)^2 - s^2/3]/(4F)$. ■

A.3 Proof of Proposition 2

(i) At m^* , the realized spread is $\mu = w + c - a$, so $(w + c - \mu)^2 = a^2$ and realized profit is $\Pi(a) = \lambda[(D - c)^2 - a^2]/(4w)$. Taking expectations,

$\mathbb{E}_a \Pi(m^*) = \lambda[(D - c)^2 - s^2/3]/(4w) = \Pi(m^*; a = 0) - \lambda s^2/(12w)$: forecast-error variance enters profit only as the subtraction $\lambda \sigma_a^2/(4w)$, a pure tax. The machine operates iff $\mathbb{E}_a \Pi(m^*) \geq F$, i.e., iff $(D - c)^2 - s^2/3 \geq 4wF/\lambda$, i.e., iff $s^2 \leq \bar{s}^2(w) = 3[(D - c)^2 - 4wF/\lambda]$. A regime shift raising s above $\bar{s}(w)$ makes continued operation unprofitable: exit.

(ii) From Step 3 above, conditional volume is $V(a) = \lambda(D - c + a)/(2w)$, strictly increasing in a . The conditional per-trade margin is

$$m^* - c - a + \mathbb{E}[\varepsilon \mid \text{accept}] = w - a + \frac{\tau_D - w}{2} = \frac{D - c - a}{2},$$

strictly decreasing in a . Hence $\text{Cov}(V, a) = \frac{\lambda}{2w} \text{Var}(a) = \lambda s^2/(6w) > 0$, and realized profit $\Pi(a) = V(a) \times (D - c - a)/2 = \lambda[(D - c)^2 - a^2]/(4w)$ is minimized at $|a| = s$: the machine's highest-volume states ($a = s$) are its lowest-profit states.

(iii) $\partial \bar{s}^2 / \partial w = -12F/\lambda < 0$. ■

A.4 Proof of Corollary 1

Firm j operates iff $s_j \leq \bar{s}$ (Proposition 2(i), holding the segment fixed). After the shock, firm j 's error scale is κs_j , so it operates iff $\kappa s_j \leq \bar{s}$, i.e., iff $s_j \leq \bar{s}/\kappa$. The shock therefore removes exactly the firms with $s_j \in (\bar{s}/\kappa, \bar{s}]$: those that operated before but whose forecast quality is too low to survive the scaling. Firms with $s_j \leq \bar{s}/\kappa$ remain. ■

A.5 Proof of Proposition 3

Projection MSEs. Under the diffuse prior, the least-squares estimate of v from k household comps is the comp mean $\hat{v}_0 = \bar{p} = v + \bar{\xi}$, with error variance $V_0 = \sigma_\xi^2/k$. Adding the de-spread machine signal $v + a$ with variance σ_a^2 , precision weighting gives

$$\hat{v}_1 = V_1 \left(\frac{k\bar{p}}{\sigma_\xi^2} + \frac{v + a}{\sigma_a^2} \right), \quad V_1 = \left(\frac{k}{\sigma_\xi^2} + \frac{1}{\sigma_a^2} \right)^{-1}.$$

Two identities follow from $V_1(k/\sigma_\xi^2 + 1/\sigma_a^2) = 1$:

$$V_0^2 \frac{k}{\sigma_\xi^2} = V_0, \quad V_1^2 \frac{k}{\sigma_\xi^2} = V_1 \left(1 - \frac{V_1}{\sigma_a^2} \right). \quad (\text{A.1})$$

Group errors and the gap. The informed buyer's error decomposes as

$$\hat{v}_1 - v = V_1 \frac{k}{\sigma_\xi^2} \bar{\xi} + \frac{V_1}{\sigma_a^2} a.$$

Conditional on the machine error a (one model, one error—common across all machine comps in the segment), the informed group's mean error is $\text{gap} = (V_1/\sigma_a^2) a$ and its within-group variance is $V_1^2 k / \sigma_\xi^2$. The uninformed group has mean error 0 and variance V_0 . (Unconditionally, $\text{Var}(\hat{v}_1 - v) = V_1^2 k / \sigma_\xi^2 + V_1^2 / \sigma_a^2 = V_1$, confirming the MSE.)

The dispersion expression. A fraction $\rho = \rho(v_m) = 1 - (1 - v_m)^k$ of buyers draw at least one machine comp. The cross-sectional variance of buyer errors is the within-group mixture plus the between-group term, $\rho(1 - \rho) \mathbb{E}_a[\text{gap}^2] = \rho(1 - \rho)(V_1^2 / \sigma_a^4) \sigma_a^2 = \rho(1 - \rho) V_1^2 / \sigma_a^2$:

$$W_B(\rho) = \rho \frac{V_1^2 k}{\sigma_\xi^2} + (1 - \rho) V_0 + \rho(1 - \rho) \frac{V_1^2}{\sigma_a^2}.$$

Substituting (A.1) and collecting terms,

$$W_B(\rho) = V_0 + \rho(V_1 - V_0) - \rho^2 \frac{V_1^2}{\sigma_a^2},$$

so $dW_B/d\rho = (V_1 - V_0) - 2\rho V_1^2 / \sigma_a^2$, linear in ρ . At $\rho = 0$ it equals $V_1 - V_0 < 0$ (the machine signal strictly reduces the projection MSE); at $\rho = 1$ it equals $V_1 - V_0 - 2V_1^2 / \sigma_a^2 < 0$. A linear function negative at both endpoints is negative on $[0, 1]$, so W_B is strictly decreasing in ρ ; since $\rho'(v_m) = k(1 - v_m)^{k-1} > 0$, W_B is strictly decreasing in v_m .

(i) In household-to-household trades, $p_i - (v + \varepsilon_i) = (\hat{v}_i - v) - \theta D \mathbf{1}\{\text{mover}\} + b_i$; the buyer-error component of the within-segment dispersion of prices around fundamentals is W_B , which falls when $v_m > 0$. The machine is not a party to any of these trades.

(ii) Exit sets ρ from $\rho^{pre} = \rho(v_m^{pre})$ to 0:

$$\Delta W_B = W_B(0) - W_B(\rho^{pre}) = \rho^{pre} \left[(V_0 - V_1) + \rho^{pre} \frac{V_1^2}{\sigma_a^2} \right] > 0,$$

the form $\rho(v_m^{pre})(V_0 - V_1 + \text{cross-terms})$ stated in the text. Its derivative in ρ^{pre} is $(V_0 - V_1) + 2\rho^{pre} V_1^2 / \sigma_a^2 > 0$, so the dispersion increase is strictly increasing in the pre-exit machine share. ■

A.6 Proof of Proposition 4

(i) *Counting deeds.* Without the machine, recorded arm’s-length volume per unit mass of houses is λ (every mover sells) plus $(1 - \lambda)\nu$ (listing non-movers): $V_{0,total} = \lambda + (1 - \lambda)\nu$. With the machine at m^* , movers with $\varepsilon_i \leq \tau_D$ sell to the machine, which resells: two deeds per intermediated house. Movers with $\varepsilon_i > \tau_D$ and listing non-movers generate one deed each. Total recorded volume is

$$\lambda [2P(\varepsilon \leq \tau_D) + P(\varepsilon > \tau_D)] + (1 - \lambda)\nu = V_{0,total} + \lambda P(\varepsilon \leq \tau_D),$$

i.e., baseline plus machine purchases, one for one. Exit removes the term $\lambda P(\varepsilon \leq \tau_D)$, which is proportional to pre-exit exposure.

(ii) *Fire sales.* Set $b = 0$ and consider the benchmark with $\hat{v}_i = v$. A non-mover’s household sale prices at fundamental, $p_i = v + \varepsilon_i$; a mover’s prices at $v + \varepsilon_i - \theta D$. A household trade executes more than t below fundamental iff it is a mover sale, for any $0 < t \leq \theta D$: the set of such trades and the set of mover sales coincide. Without the machine, the fire-sale share of household trades is $\lambda / [\lambda + (1 - \lambda)\nu]$. With the machine, movers with $\varepsilon \leq \tau_D$ exit the household pool, leaving mover mass $\lambda P(\varepsilon > \tau_D)$, so the share falls to

$$\frac{\lambda P(\varepsilon > \tau_D)}{\lambda P(\varepsilon > \tau_D) + (1 - \lambda)\nu} < \frac{\lambda}{\lambda + (1 - \lambda)\nu}.$$

Exit restores the original share. ■

B Data Appendix

B.1 iBuyer entity-name patterns

Table A1 lists the entity-name patterns applied to the first-listed buyer and seller names on each deed. Patterns are anchored to legal entities disclosed in SEC Exhibit 21.1 subsidiary lists (Opendoor, Redfin), SEC 8-K loan exhibits (Offerpad), and state registrations of the deed-holding trust (Zillow). Matching is case-insensitive on normalized names; generic strings that could collide with unrelated entities (e.g., bare “OD”) are required to appear with their state-specific suffixes.

Table A1: Entity-name patterns for iBuyer identification in deed records

Firm	Pattern family	Example deed entities
Opendoor	OPENDOOR*; OD {ARIZONA, NEVADA, TEXAS}*	Opendoor Property Trust I; Opendoor Property C LLC; OD Arizona D LLC; Opendoor Homes Phoenix 2 LLC
Zillow	ZILLOW*	Zillow Homes Property Trust
Offerpad	OFFERPAD*; OP SPE*	Offerpad (SPVBORROWER1), LLC; OP SPE PHX1 LLC; OP SPE TPA1 LLC
RedfinNow	REDFINNOW*	RedfinNow Borrower LLC

Notes: Patterns verified against SEC Ex-21.1 subsidiary filings and state corporate registrations. The scan classifies a deed as an iBuyer purchase when the pattern matches the first-listed buyer and as a disposition when it matches the first-listed seller.

B.2 Sample filter waterfall

Table A2 reports the deed counts surviving each filter, accumulated over the 22 panel states and 2010Q1–2024Q2.

Table A2: Sample construction: filter waterfall (22 states, 2010Q1–2024Q2)

Filter stage	Records
All deed records, 2010Q1–2024Q2	93,909,691
Arm’s-length market sale (category A)	50,187,509
Residential indicator	49,494,349
Single-family or condominium	48,756,024
Price \$10,000–\$10 million	43,577,709
Excl. inter-family transfers	42,535,591
Excl. foreclosure/REO	42,535,581
Valid 5-digit ZIP (final sample)	42,382,121

B.3 Hedonic model fit

The hedonic specification (log price on log square footage, bedrooms, bathrooms, age, age squared, condo indicator, and ZIP fixed effects, estimated

yearly by state on non-iBuyer trades; bedrooms and bathrooms carry missing-value indicators because some states’ assessors do not report room counts) attains R^2 in the range 0.34–0.85 across state-years, with a mean of 0.60. Residual moments by state-year, and sensitivity of the dispersion outcomes to adding lot size and excluding new construction, are available in the replication package.

C Balance, calibration, and additional robustness

C.1 Pre-treatment balance

Table A3 reports pre-treatment characteristics across Zillow-exposure groups. The pattern is the one the design needs: exposed ZIPs differ from controls on housing-stock composition (newer, more single-family, more homogeneous in size, higher continuing-machine exposure)—the sorting of Proposition 1—but the normalized differences on 2019–2021Q3 price appreciation and 2019–2020 volume growth are small, so the froth-driven mean-reversion channel that would confound the exit estimate has little purchase on observables.

C.2 Calibration of the price-discovery bound

Table A4 reports the model’s predicted exit effect on dispersion across comp-window sizes and buyer-error shares, with σ_a pinned from Zillow’s own purchase-price residual dispersion. Every cell lies below the data’s one-sided ceiling; the test caps the externality rather than confirming it.

C.3 Donut specification and replication

The donut specification referenced in Section 6.2 re-estimates equation (18) excluding both the inventory-liquidation quarters and Zillow’s 2021Q1–2021Q3 buying ramp, so the pre-period reflects normal machine operation (2019Q1–2020Q4); estimates are in `out_dir/exit_donut.rds`. Replication code for every exhibit, including the 999 permutation draws, the wild cluster bootstrap, the balance and vintage tables, and the per-state hedonic fits, is in the project’s `scripts/R/` directory (scripts 01–09).

Table A3: Exit design: pre-treatment balance across Zillow-exposure groups

	Group means				Norm. diff. vs T3	
	Zero	T1	T2	T3 (high)	T1	Zero
Log median price, 2019	12.80	12.69	12.66	12.60	-0.19	-0.36
Transactions, 2019	289	627	677	718	0.23	1.19
Log price growth, 2019–2021Q3	0.27	0.27	0.28	0.29	0.14	0.09
Log volume growth, 2019–2020	0.01	0.00	0.00	-0.01	-0.09	-0.12
Residual SD, 2019	0.37	0.34	0.30	0.28	-0.42	-0.65
Cash share, 2019	0.26	0.27	0.24	0.23	-0.46	-0.29
Investor share, 2019	0.00	0.00	0.00	0.00	NaN	NaN
Opendoor/Offerpad exposure (%)	0.16	0.64	1.16	1.37	0.61	1.13
Stock SD log sqft	0.41	0.38	0.37	0.36	-0.27	-0.64
Share built post-1990	0.28	0.28	0.40	0.49	0.72	0.82
Single-family share	0.85	0.83	0.89	0.91	0.46	0.30
Residential parcels	4359	7872	7649	7653	-0.05	0.82
ZIPs	940	495	495	495		

Notes: ZIPs in the exit sample (CBSAs with ≥ 50 Zillow purchases), grouped by terciles of positive Zillow exposure (2018Q2–2021Q3 purchase share), with zero-exposure ZIPs separate. Pre-treatment characteristics measured in 2019 or from the housing stock. Final columns: Imbens–Rubin normalized differences between the top tercile and the comparison group ($|\cdot| < 0.25$ conventionally balanced).

Table A4: Calibrated price-discovery effect vs. the data’s ceiling

Comp window	Buyer-error share of residual variance		
	0.10	0.33	1.00
$k = 4$	0.009	0.055	0.226
$k = 6$	0.013	0.082	0.338
$k = 8$	0.017	0.109	0.449

Data: 95% one-sided ceiling (IQR measure) = 0.17 log points

Pinned σ_a (Zillow within-ZIP-year residual SD) = 0.188; published off-market error ≈ 0.10

Notes: Each cell is the model’s predicted rise in within-segment residual dispersion (log points, at mean Zillow exposure) when the machine signal is removed, $\rho(v_m)(V_0 - V_1)$ converted to a standard-deviation change, evaluated at the machine’s deed footprint ($v_m \approx$ twice the mean purchase share) and σ_a pinned from Zillow’s own purchase-price residual dispersion. The data’s 95% one-sided ceiling (0.17 log points) lies *below* the model’s most aggressive cells (large comp windows and high buyer-error share; 3 of 9 columns shown), which the design therefore rejects, and above its central calibrations, with which it is consistent. The externality is bounded tightly, not point-identified; either way it is economically small.